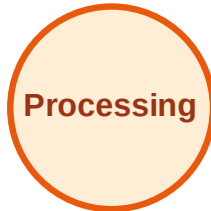


BFS Traversal

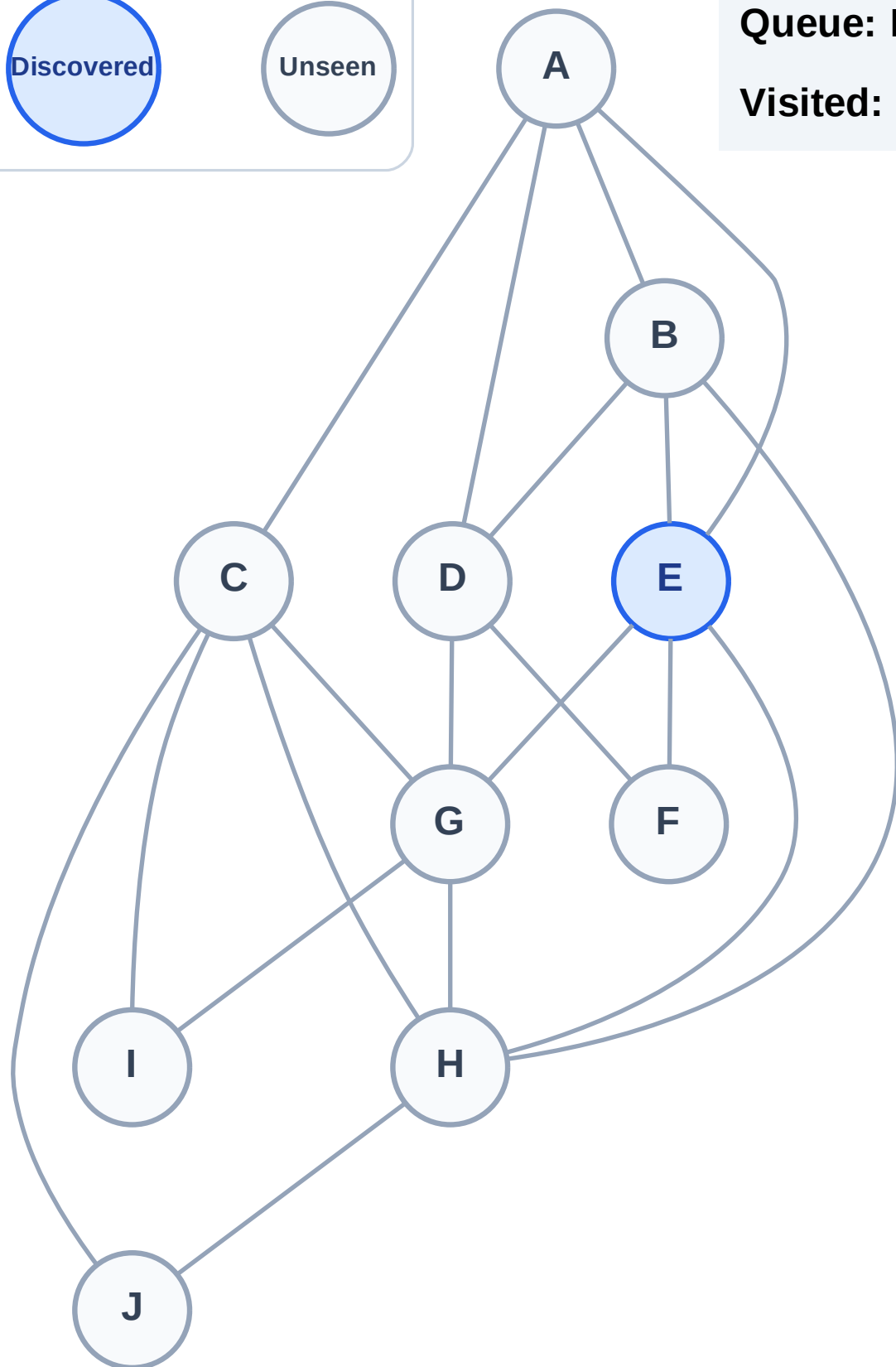
Step 1: Start at E

Legend



Queue: E

Visited: (none)



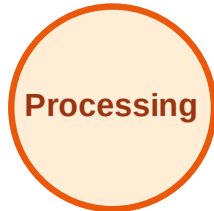
BFS Traversal

Step 2: Process node E

Legend



Complete



Processing



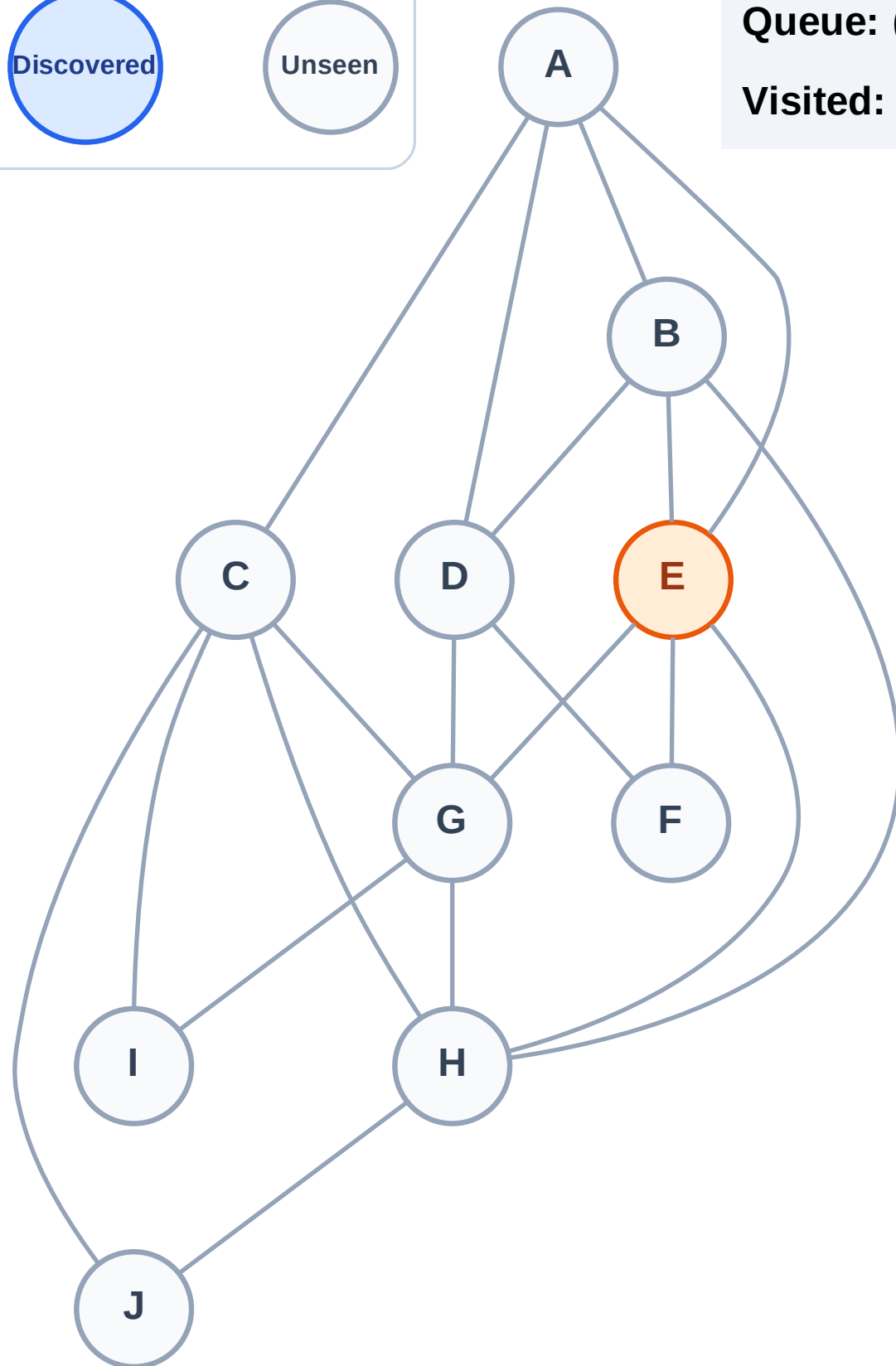
Discovered



Unseen

Queue: (empty)

Visited: (none)



BFS Traversal

Step 3: Discover A, B, F, G, H from E

Legend

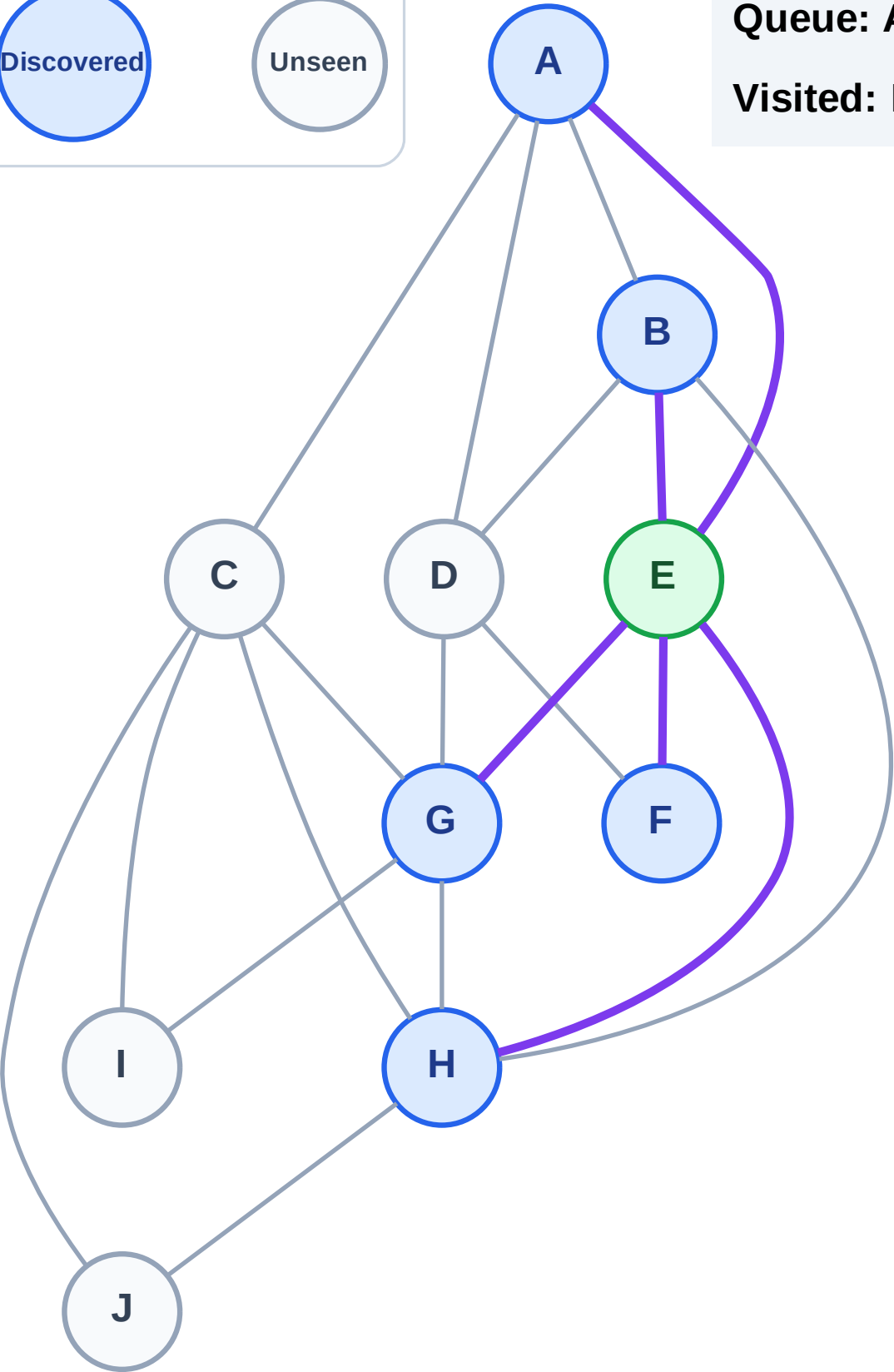
Complete

Processing

Discovered

Unseen

Queue: A → B → F → G → H
Visited: E



BFS Traversal

Step 4: Process node A

Legend

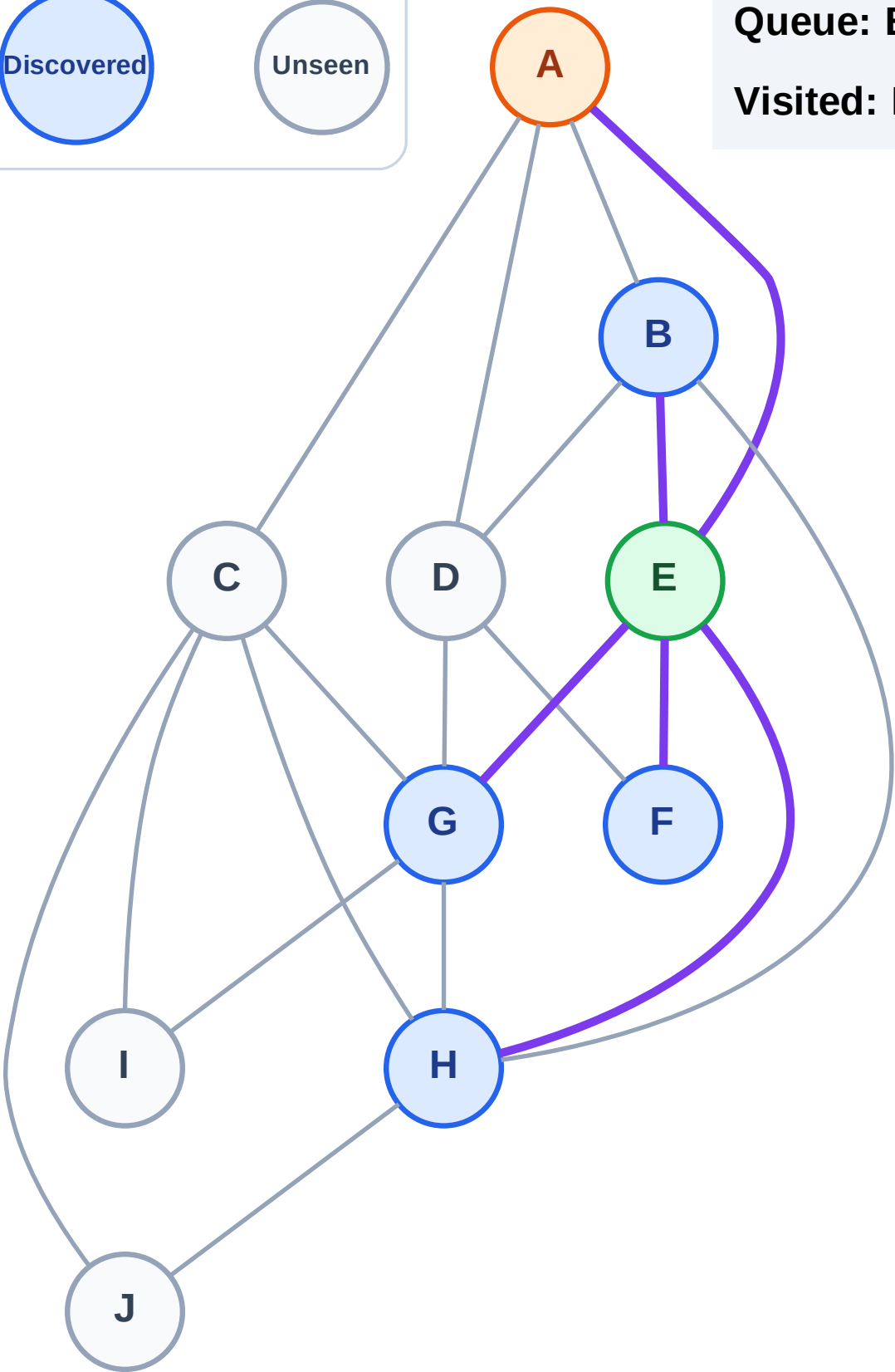
Complete

Processing

Discovered

Unseen

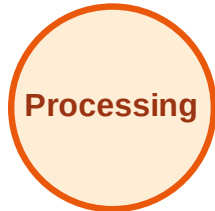
Queue: B → F → G → H
Visited: E



BFS Traversal

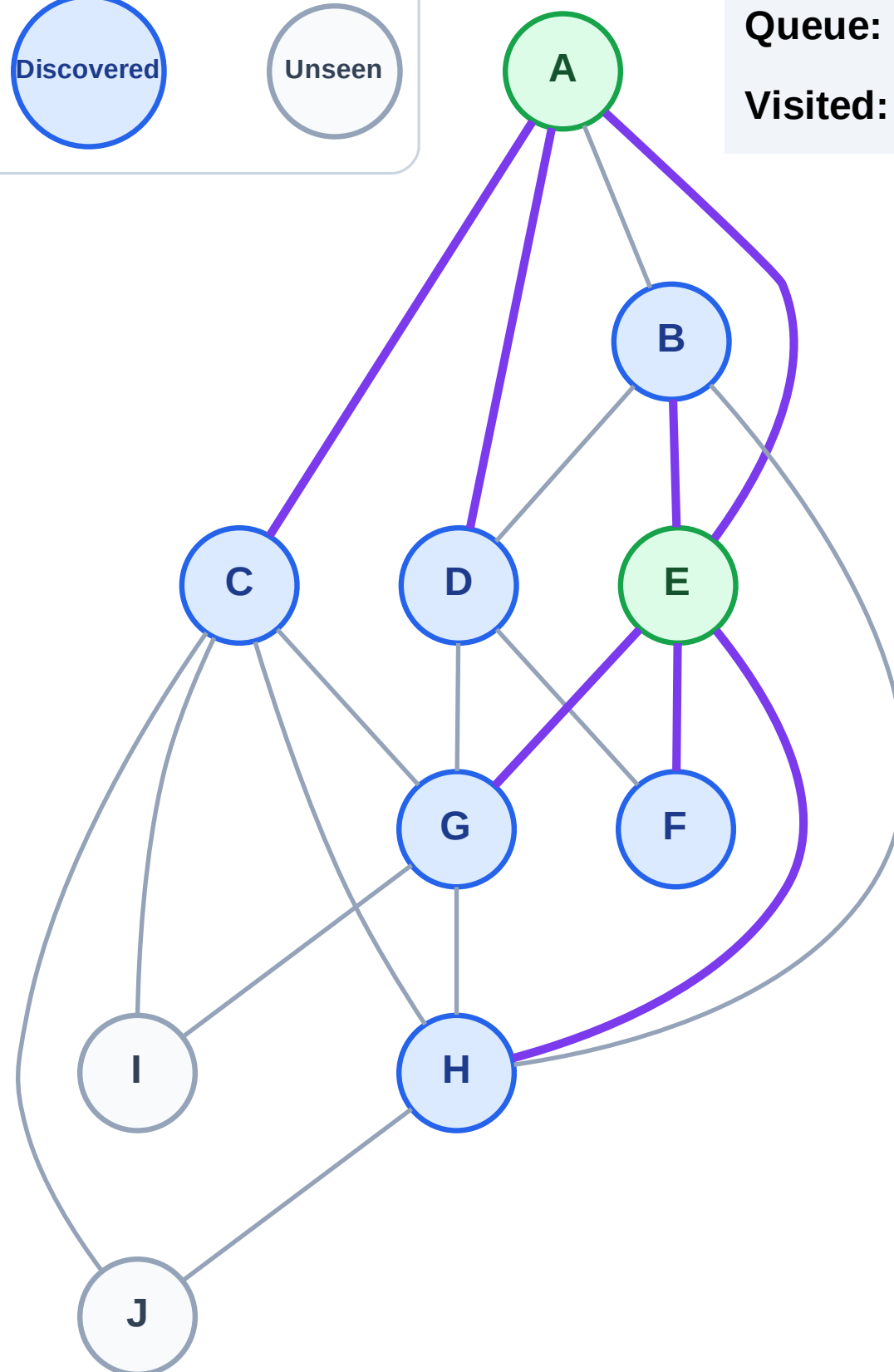
Step 5: Discover C, D from A

Legend



Queue: B → F → G → H → C → D

Visited: A, E



BFS Traversal

Step 6: Process node B

Legend

Complete

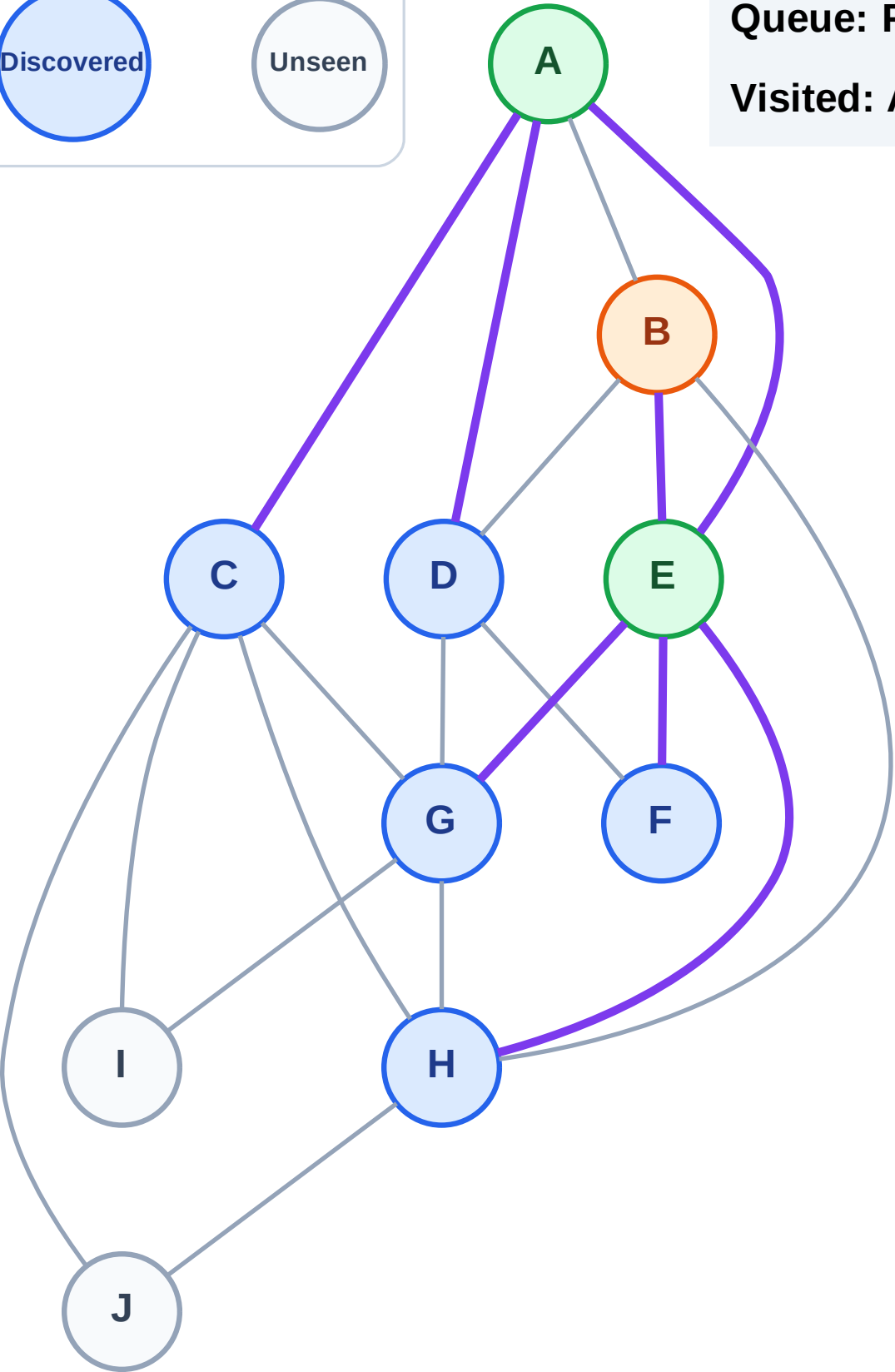
Processing

Discovered

Unseen

Queue: F → G → H → C → D

Visited: A, E



BFS Traversal

Step 7: No new neighbors from B

Legend

Complete

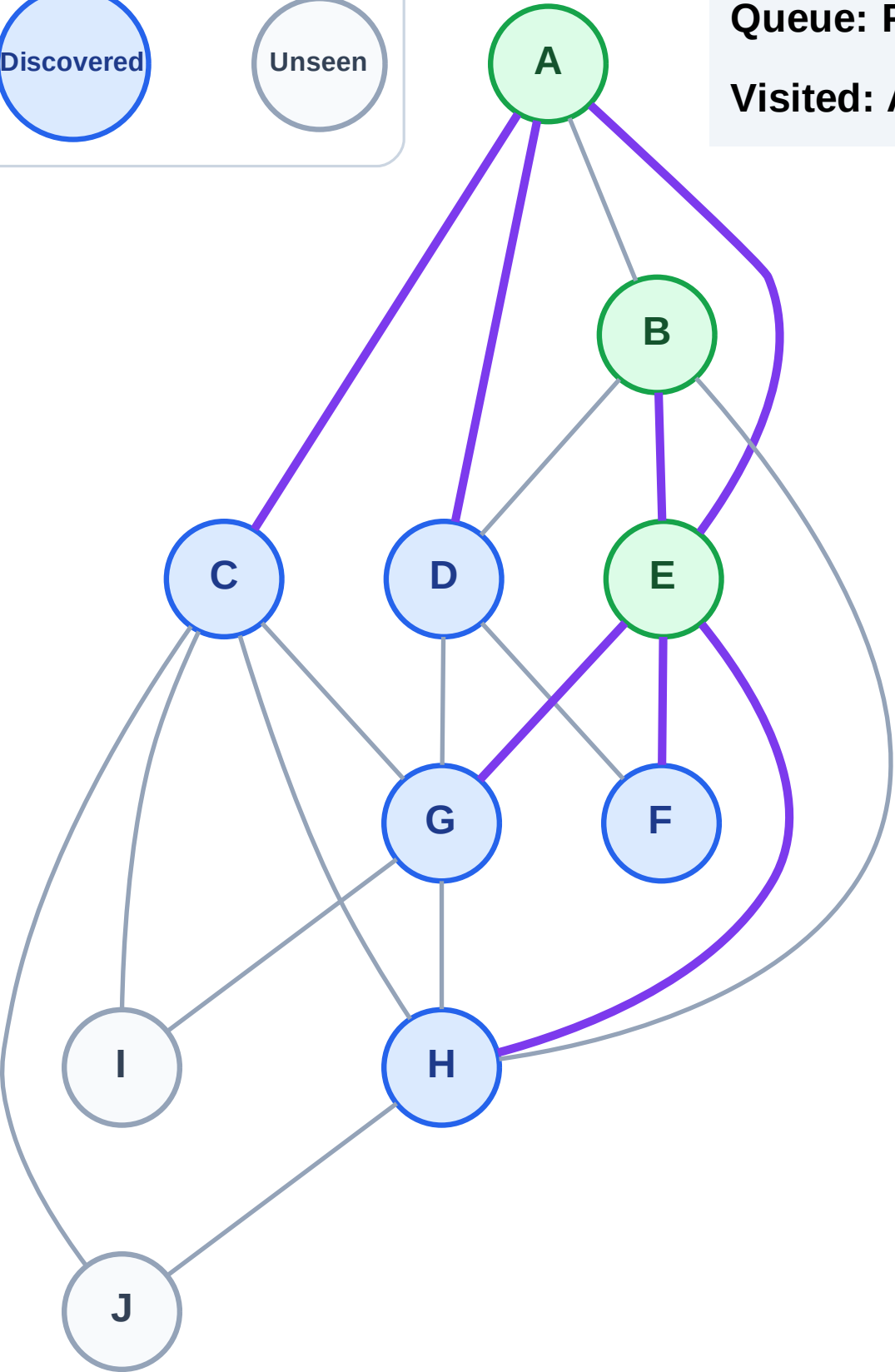
Processing

Discovered

Unseen

Queue: F → G → H → C → D

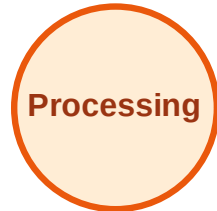
Visited: A, B, E



BFS Traversal

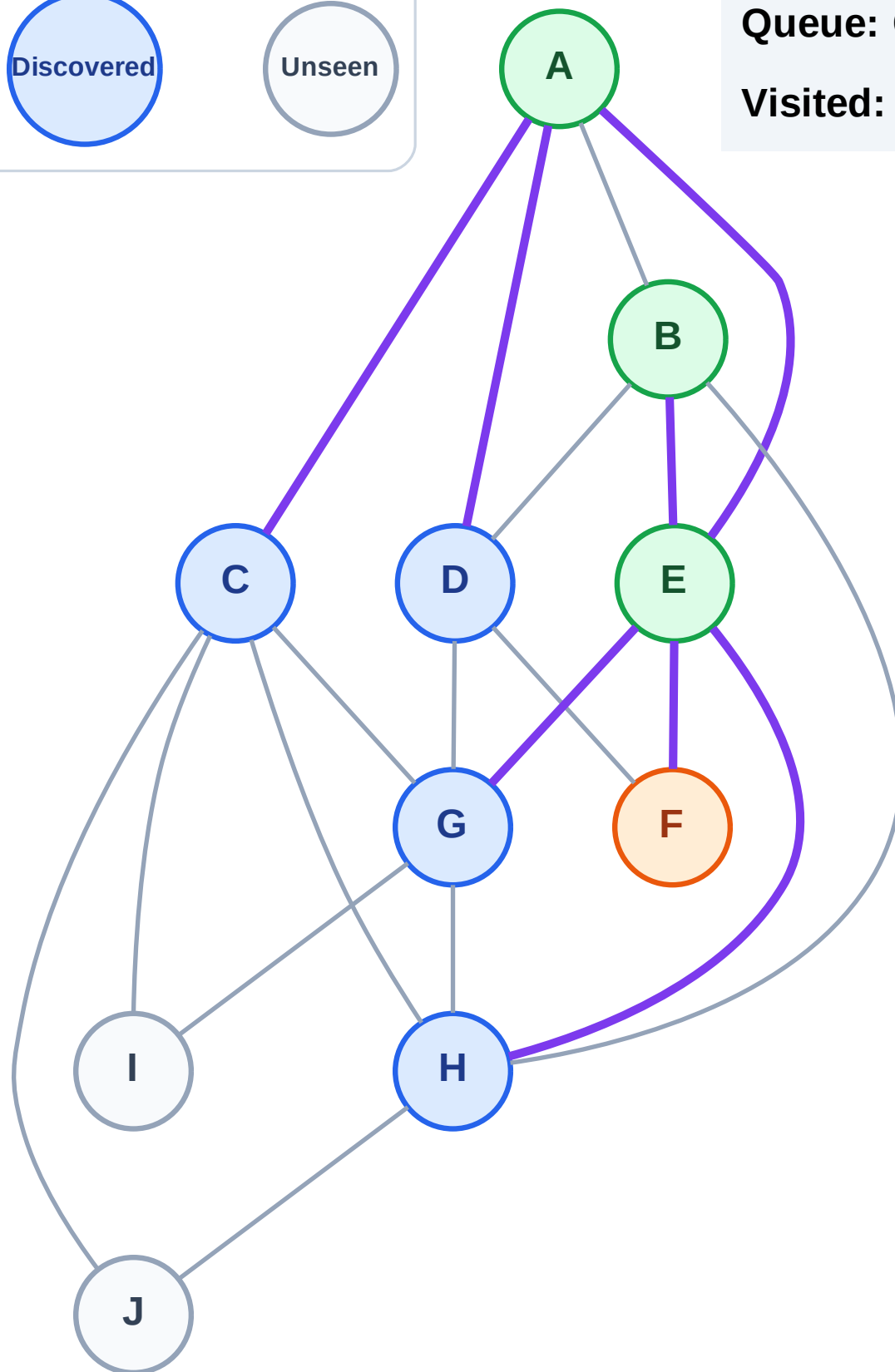
Step 8: Process node F

Legend



Queue: G → H → C → D

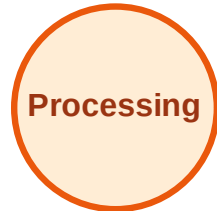
Visited: A, B, E



BFS Traversal

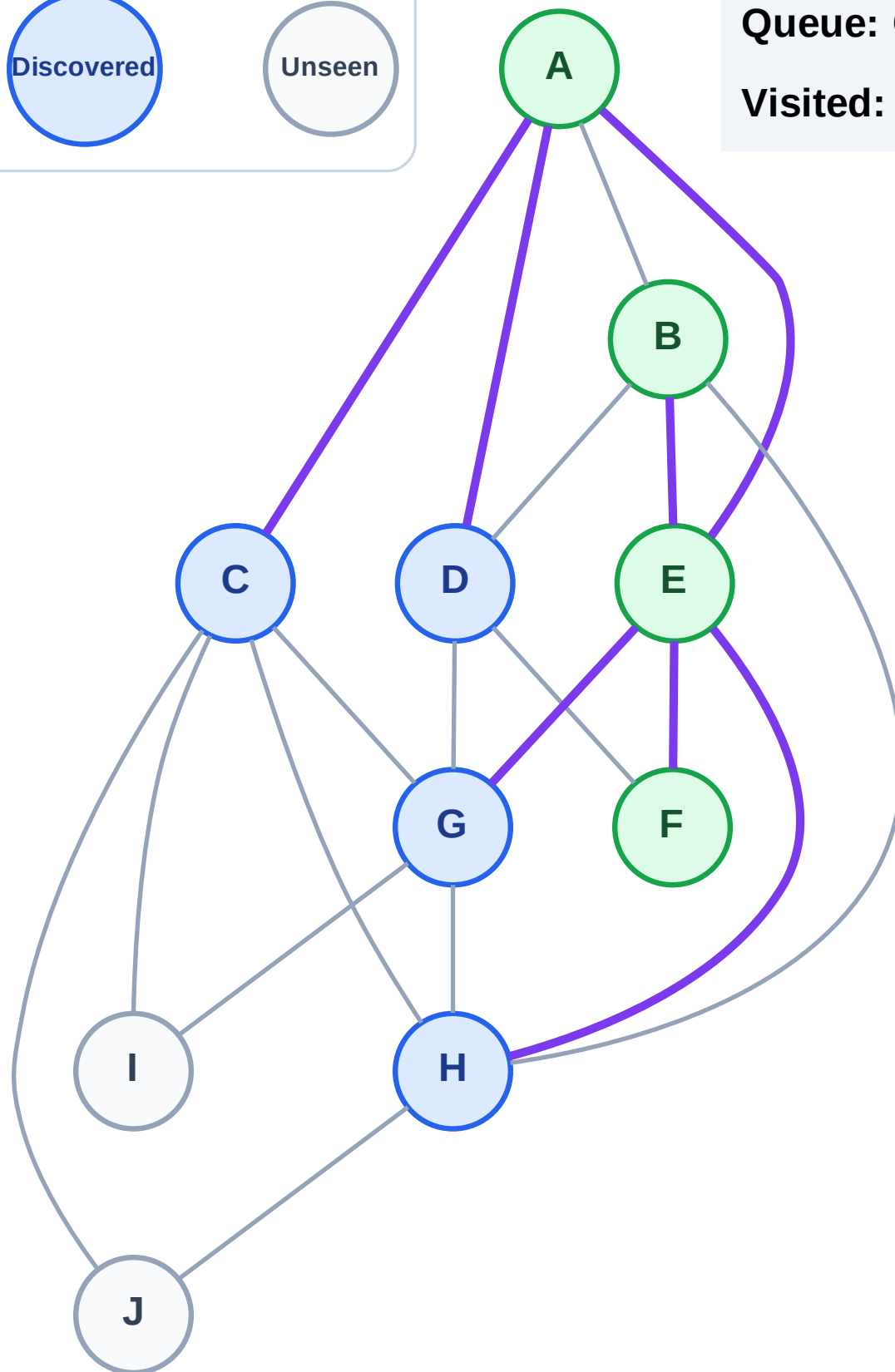
Step 9: No new neighbors from F

Legend



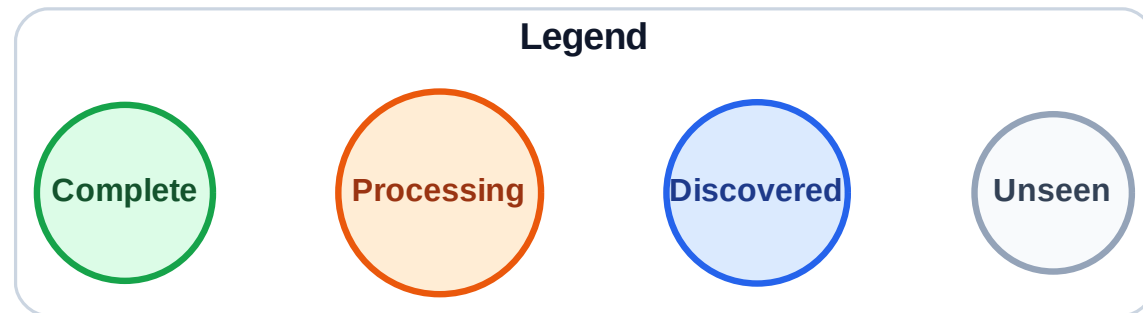
Queue: G → H → C → D

Visited: A, B, E, F

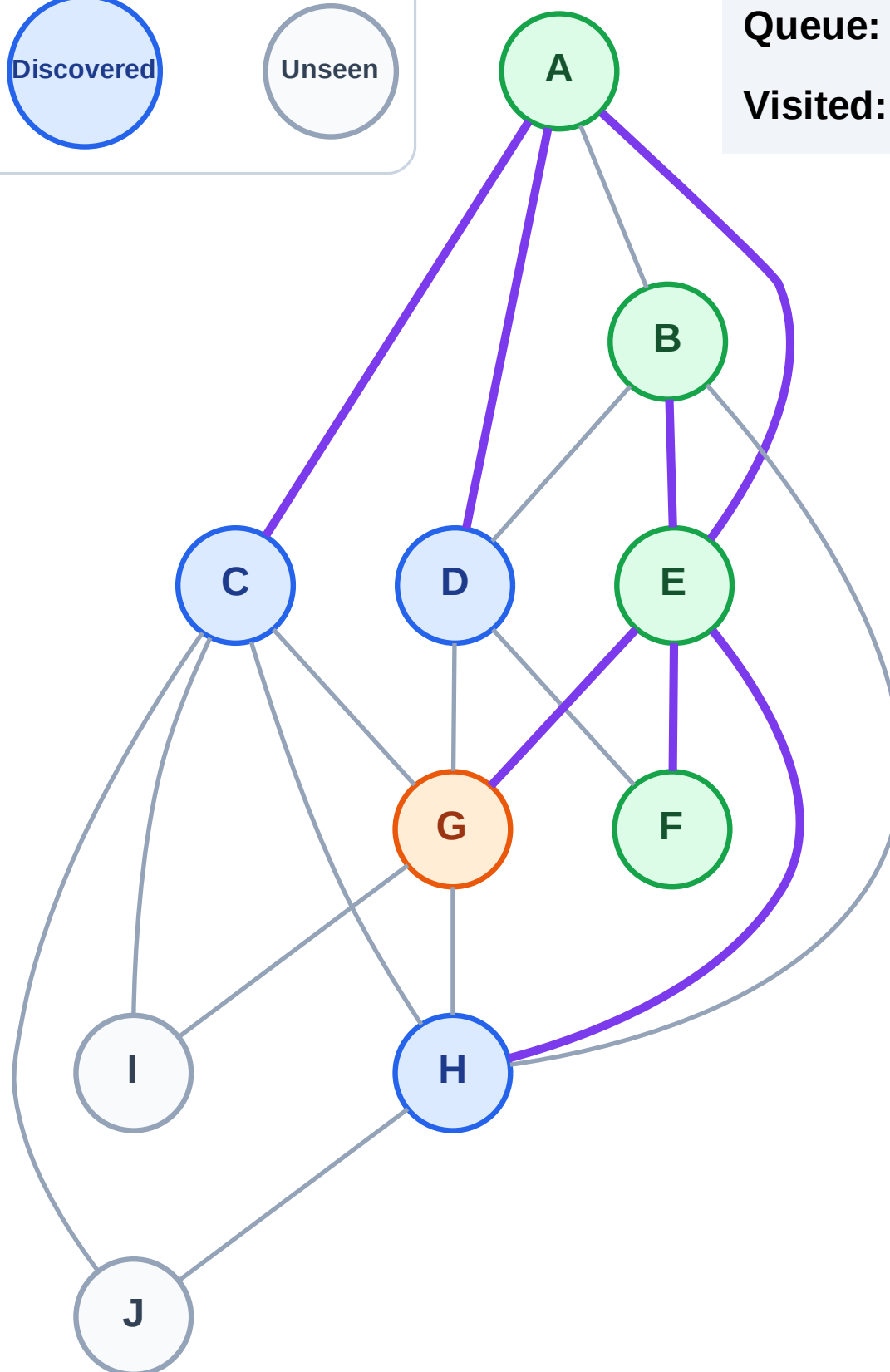


BFS Traversal

Step 10: Process node G



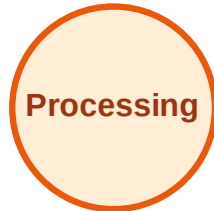
Queue: H → C → D
Visited: A, B, E, F



BFS Traversal

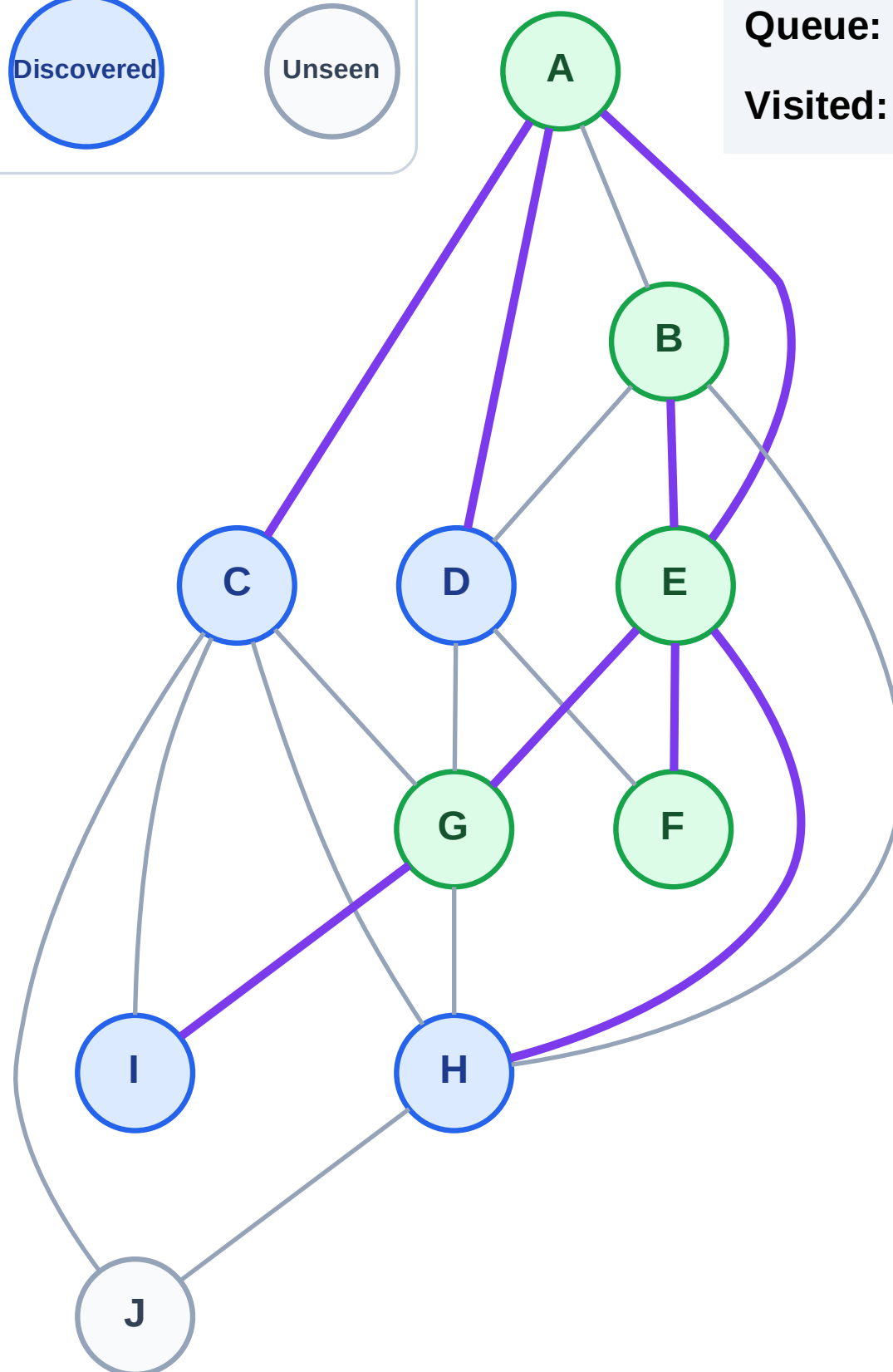
Step 11: Discover I from G

Legend



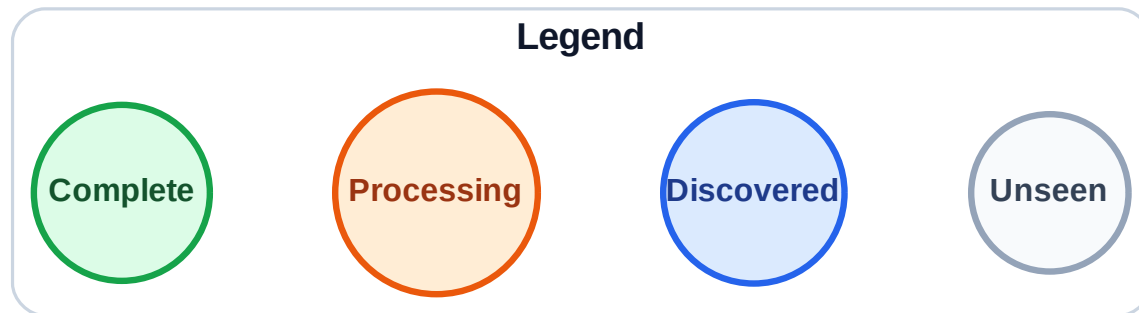
Queue: H → C → D → I

Visited: A, B, E, F, G

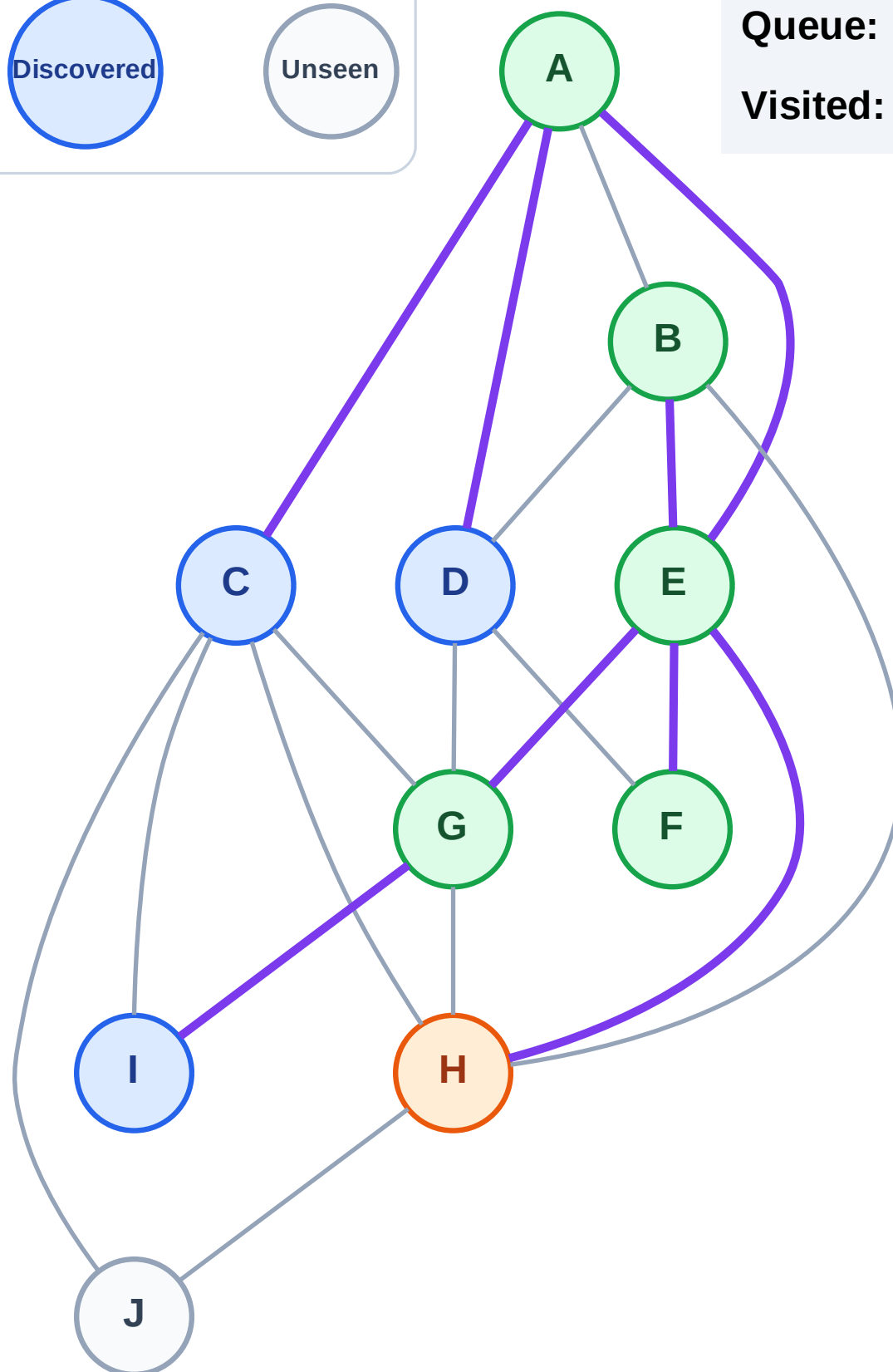


BFS Traversal

Step 12: Process node H



Queue: C → D → I
Visited: A, B, E, F, G



BFS Traversal

Step 13: Discover J from H

Legend

Complete

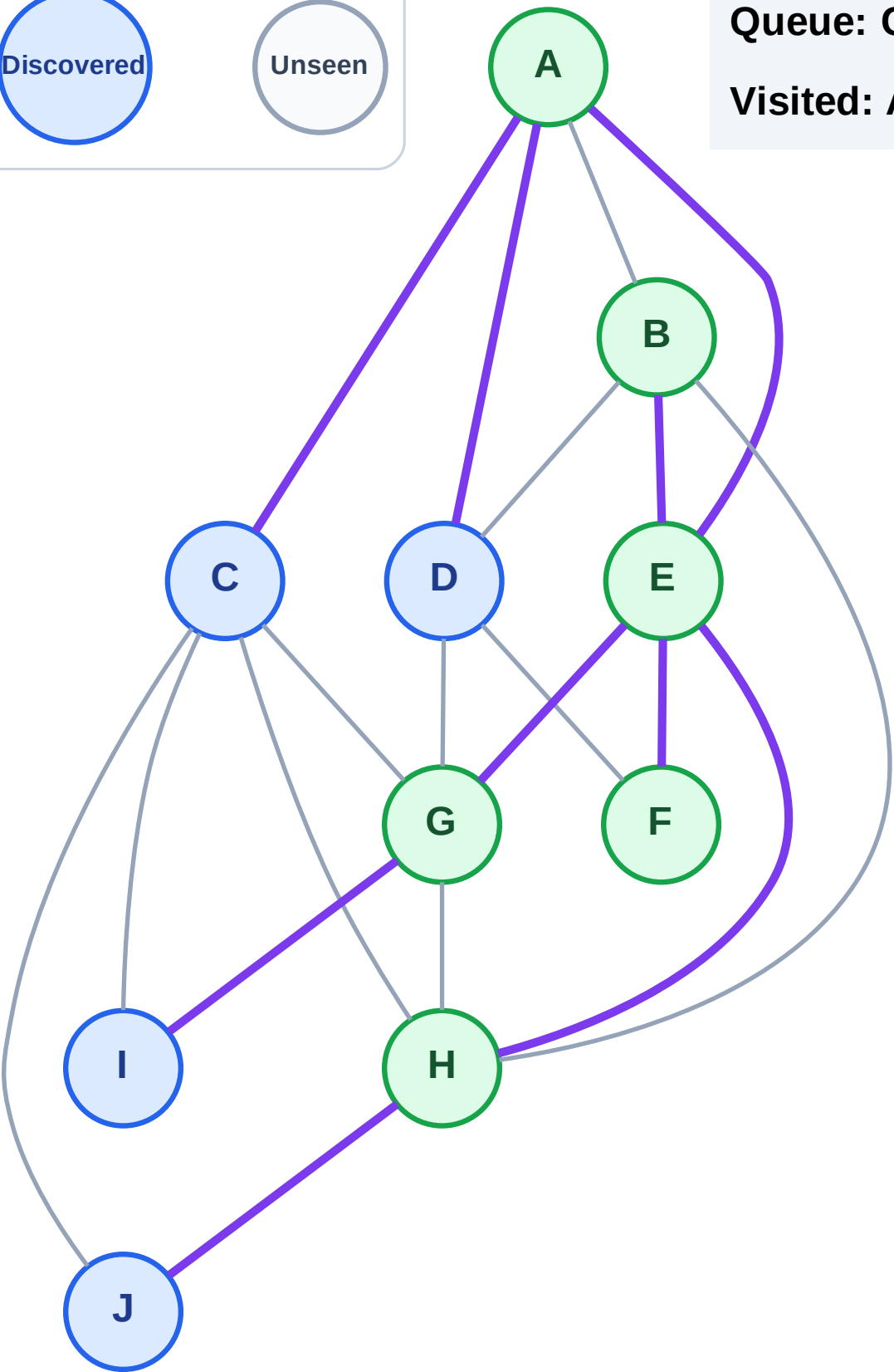
Processing

Discovered

Unseen

Queue: C → D → I → J

Visited: A, B, E, F, G, H



BFS Traversal

Step 14: Process node C

Legend

Complete

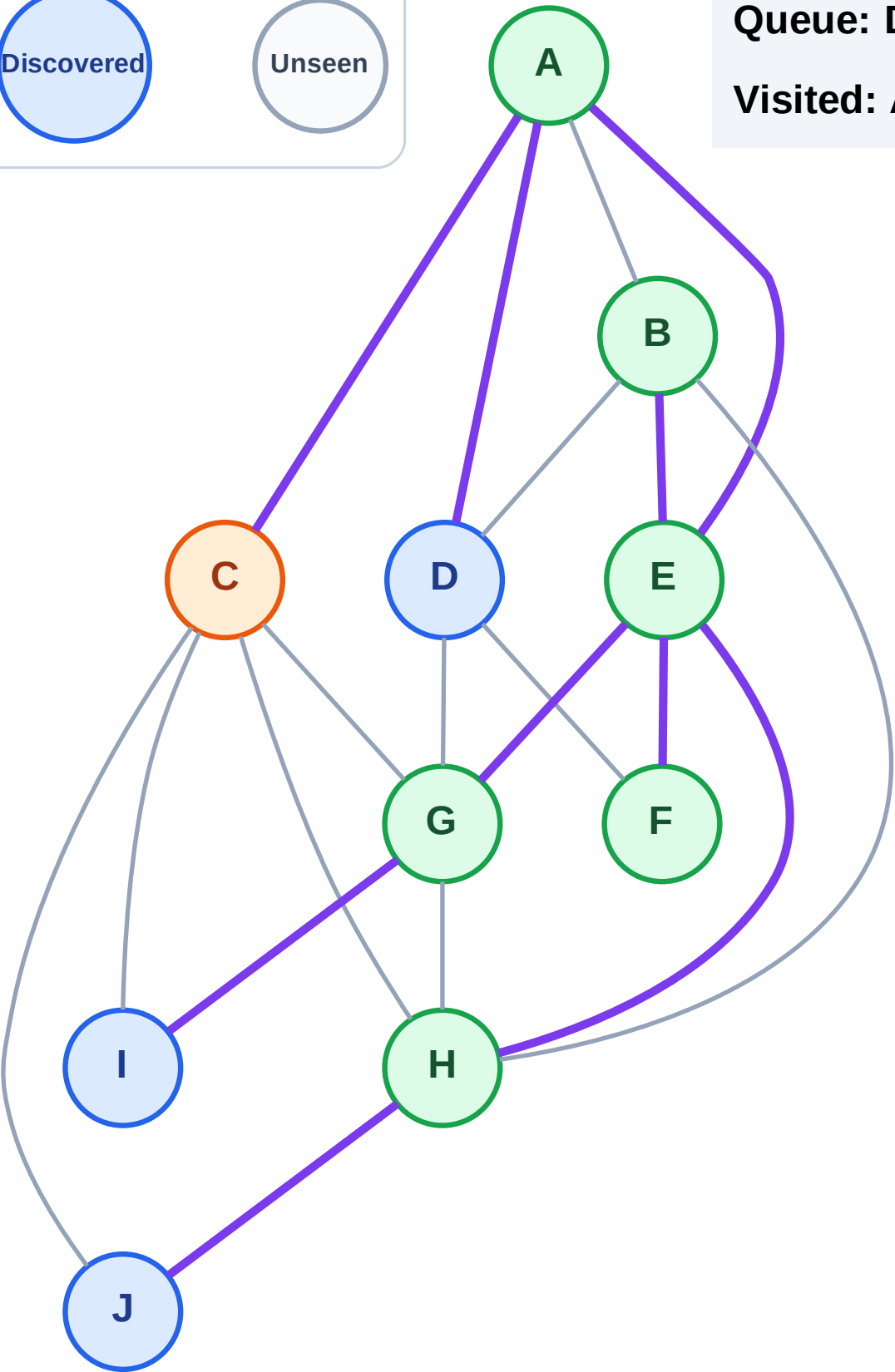
Processing

Discovered

Unseen

Queue: D → I → J

Visited: A, B, E, F, G, H



BFS Traversal

Step 15: No new neighbors from C

Legend

Complete

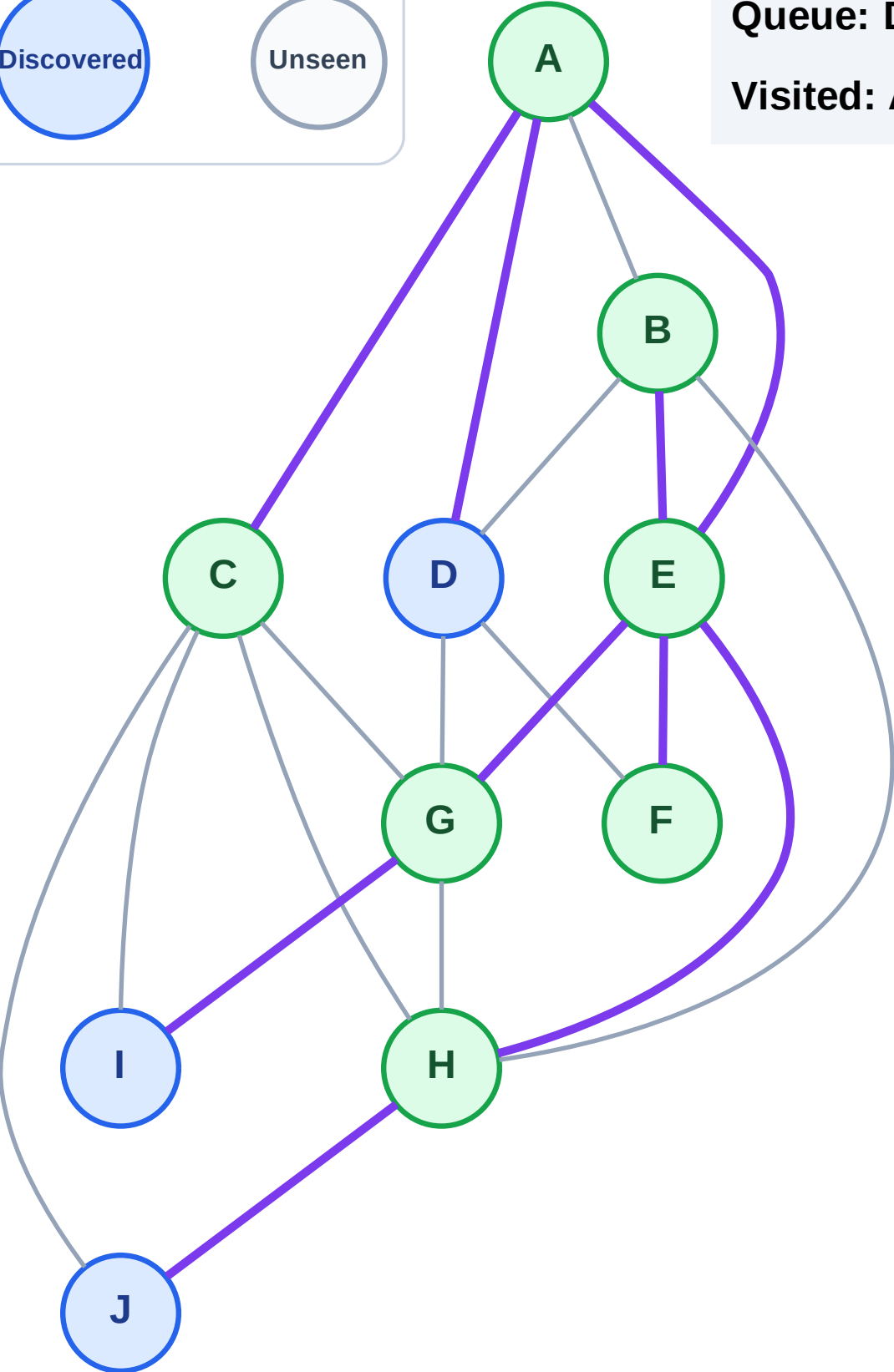
Processing

Discovered

Unseen

Queue: D → I → J

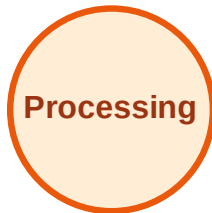
Visited: A, B, C, E, F, G, H



BFS Traversal

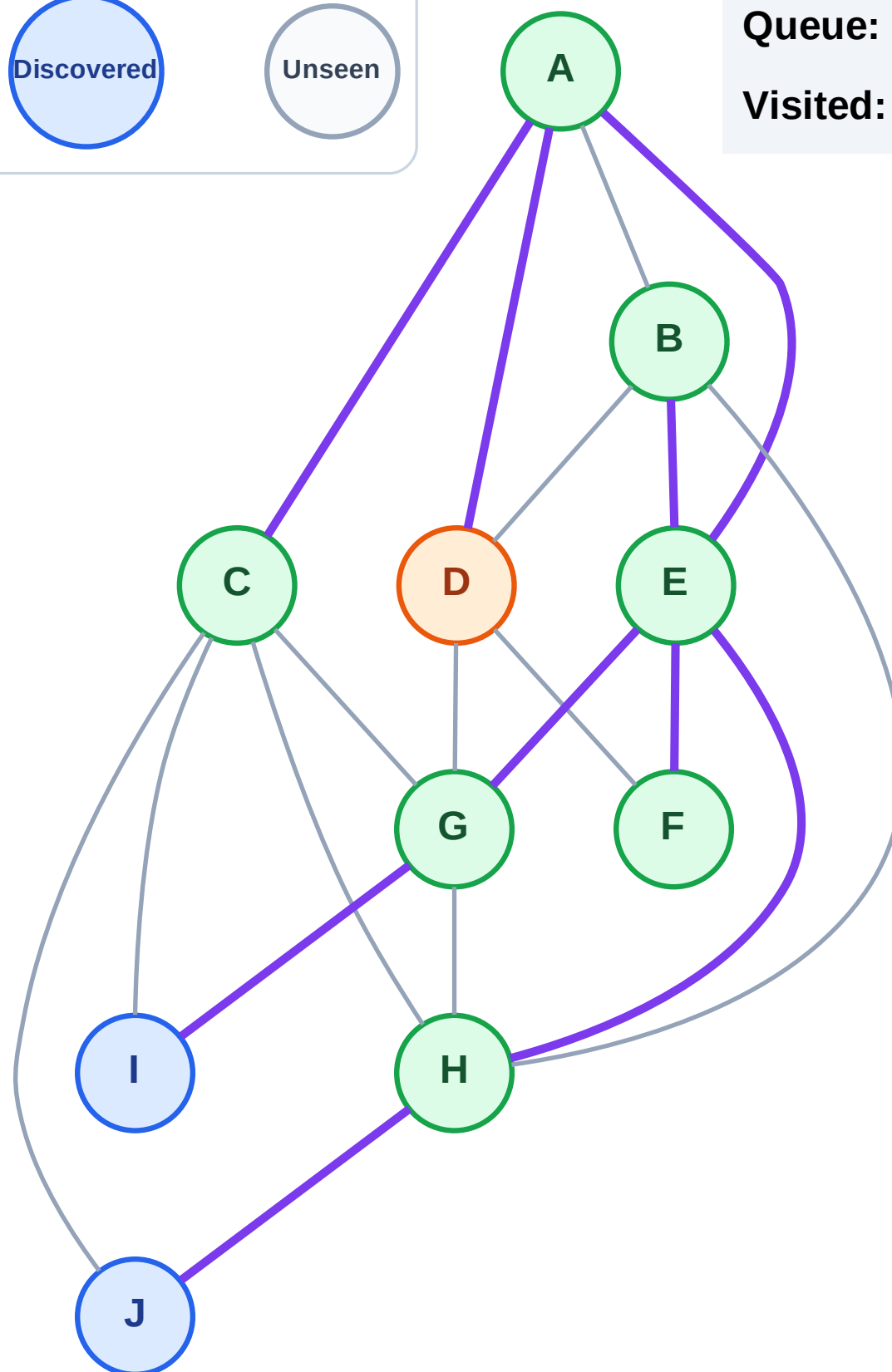
Step 16: Process node D

Legend



Queue: I → J

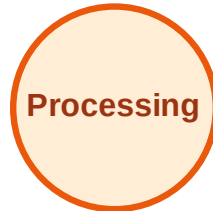
Visited: A, B, C, E, F, G, H



BFS Traversal

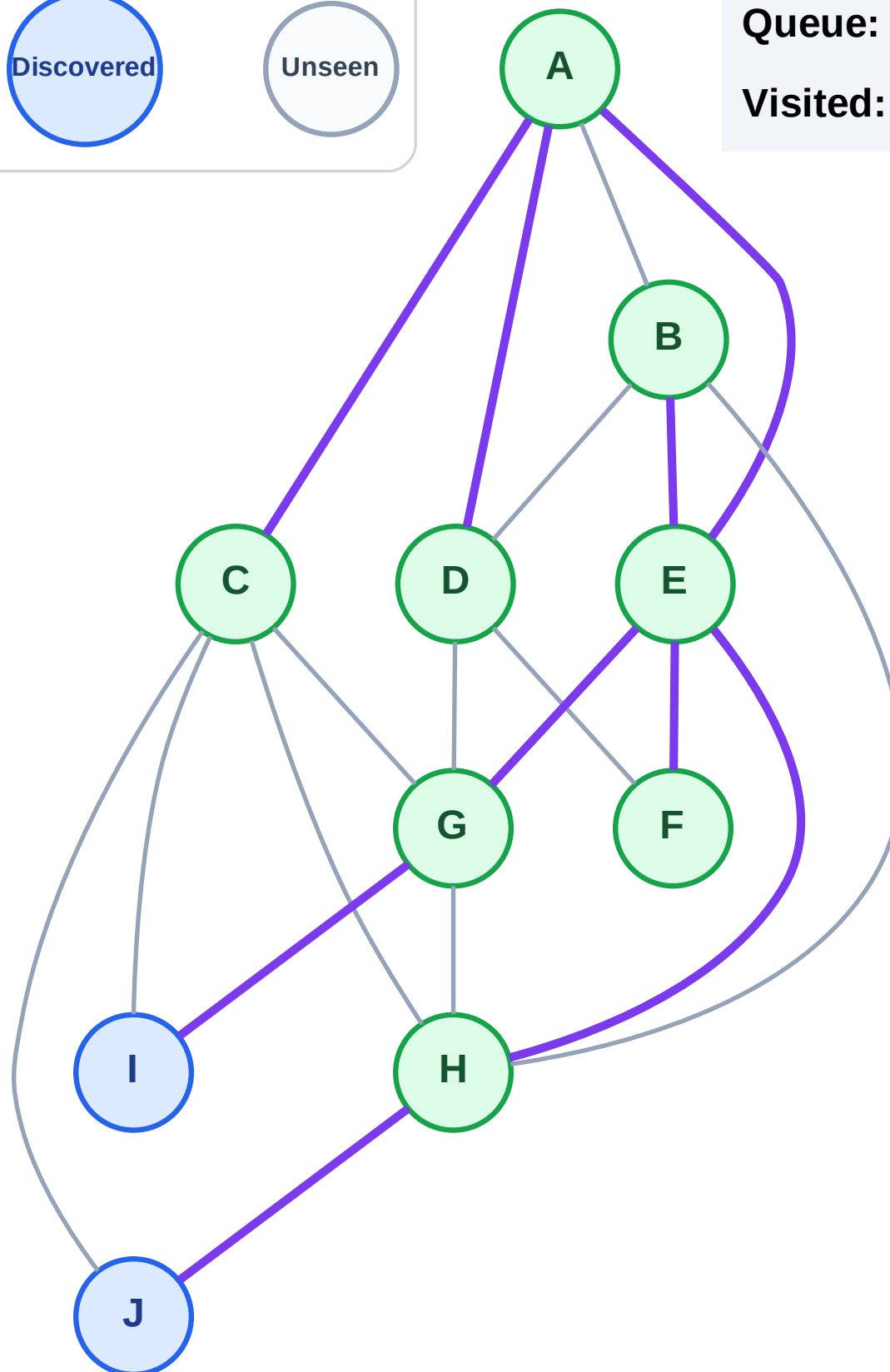
Step 17: No new neighbors from D

Legend



Queue: I → J

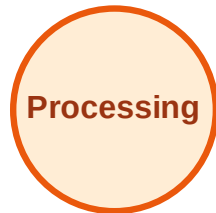
Visited: A, B, C, D, E, F, G, H



BFS Traversal

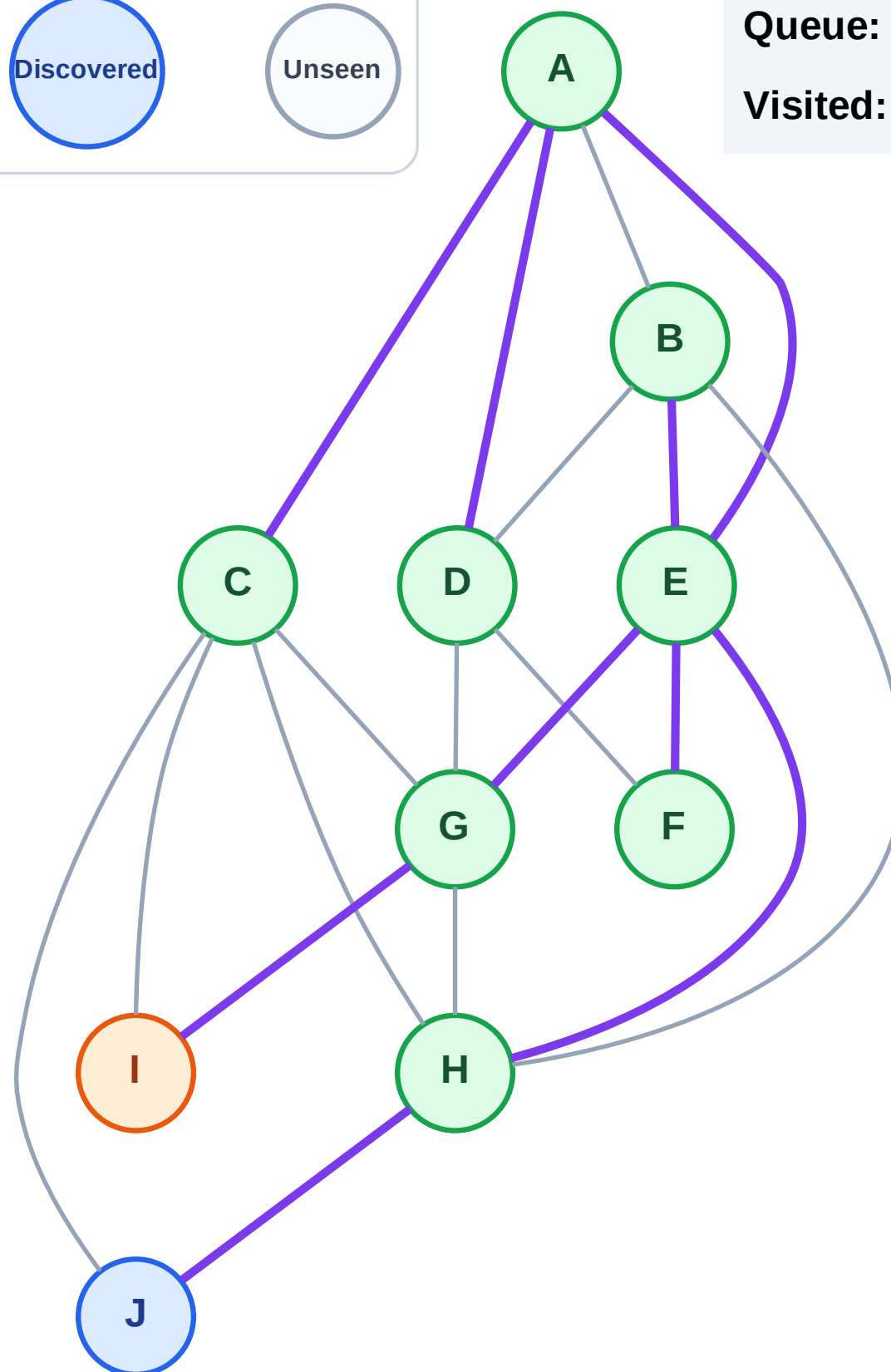
Step 18: Process node I

Legend



Queue: J

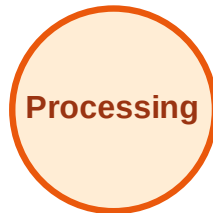
Visited: A, B, C, D, E, F, G, H



BFS Traversal

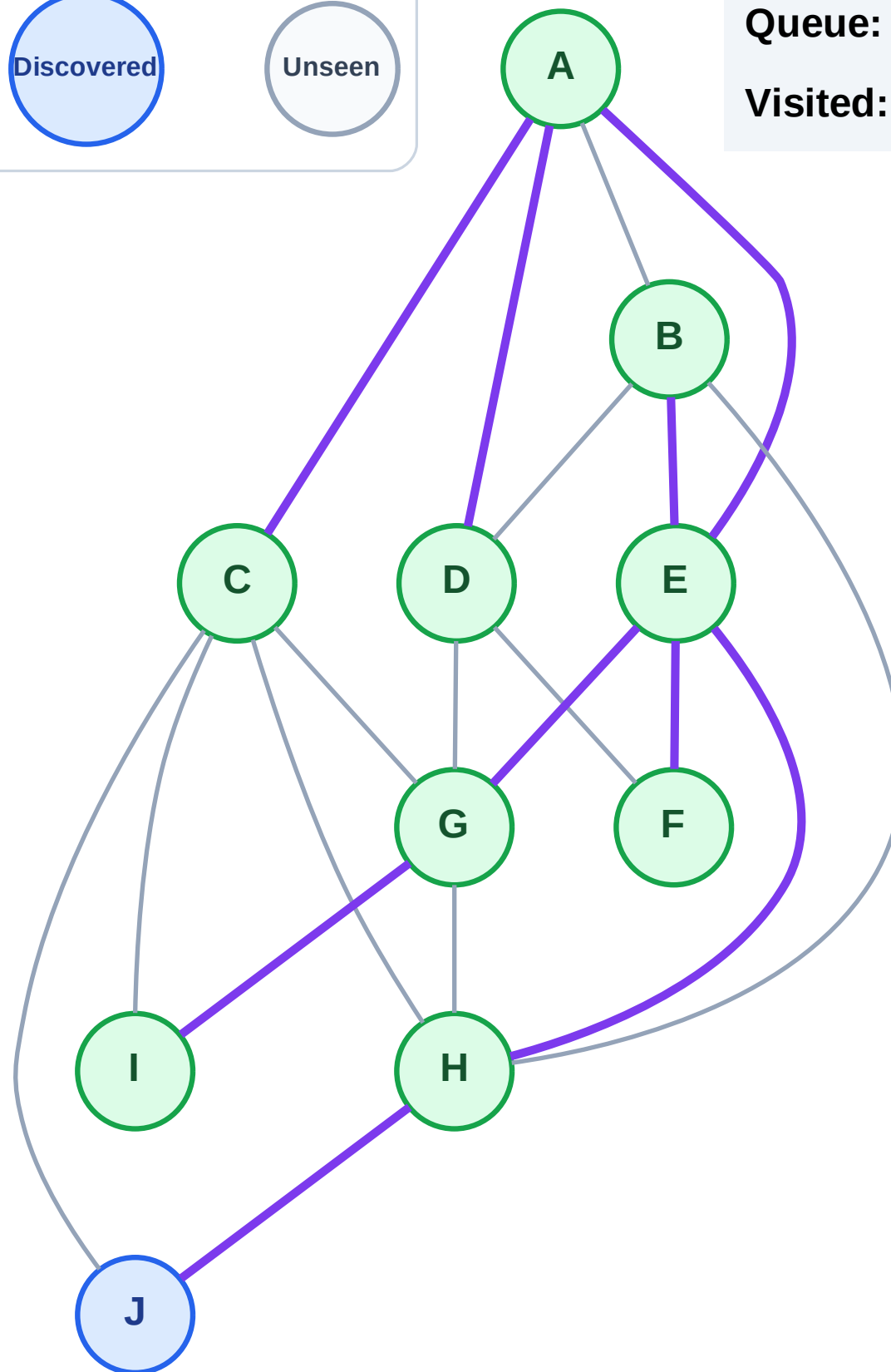
Step 19: No new neighbors from I

Legend



Queue: J

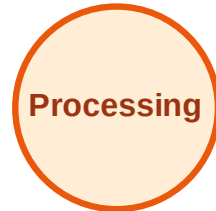
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

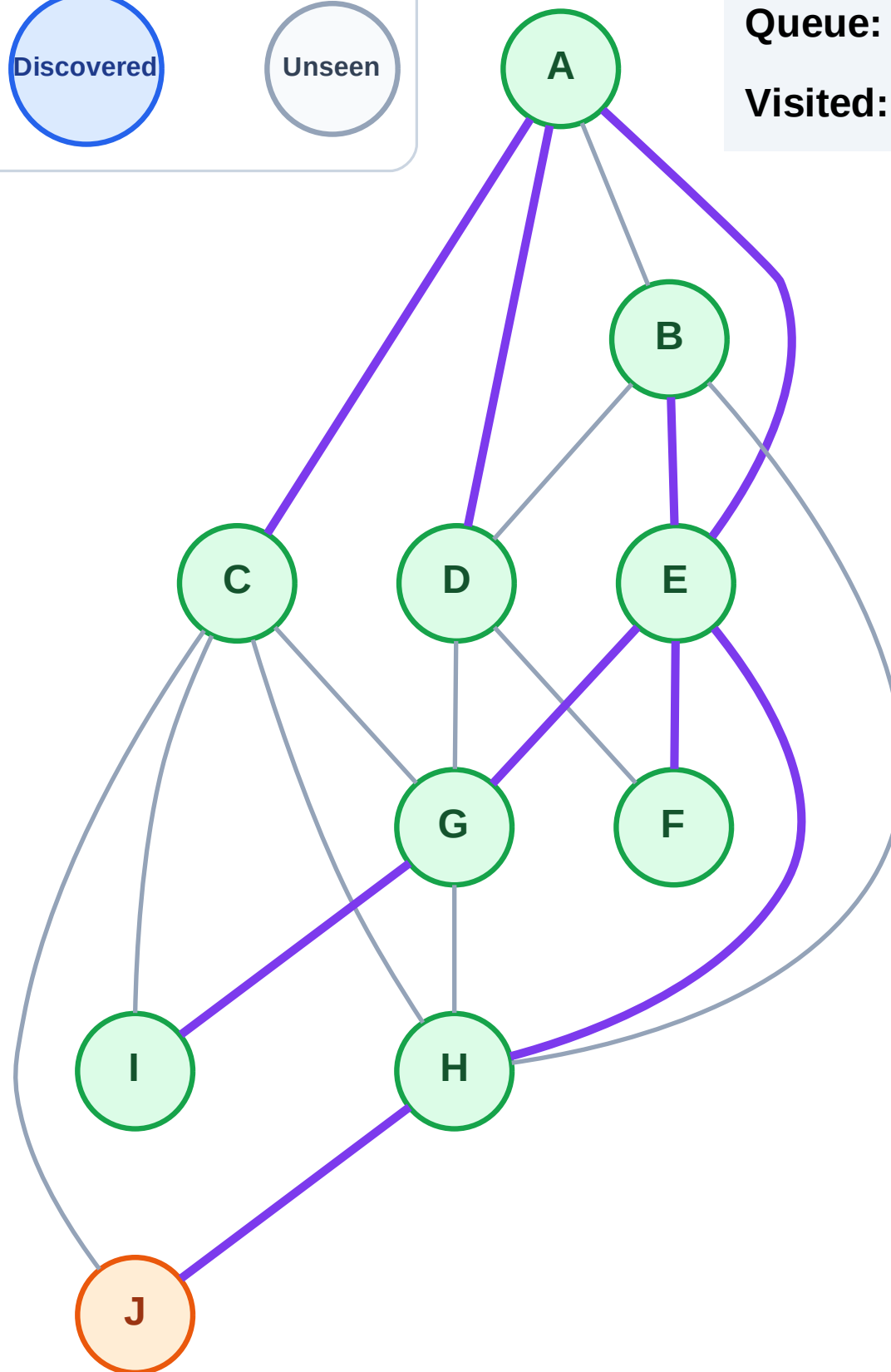
Step 20: Process node J

Legend



Queue: (empty)

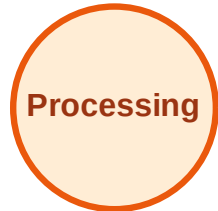
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

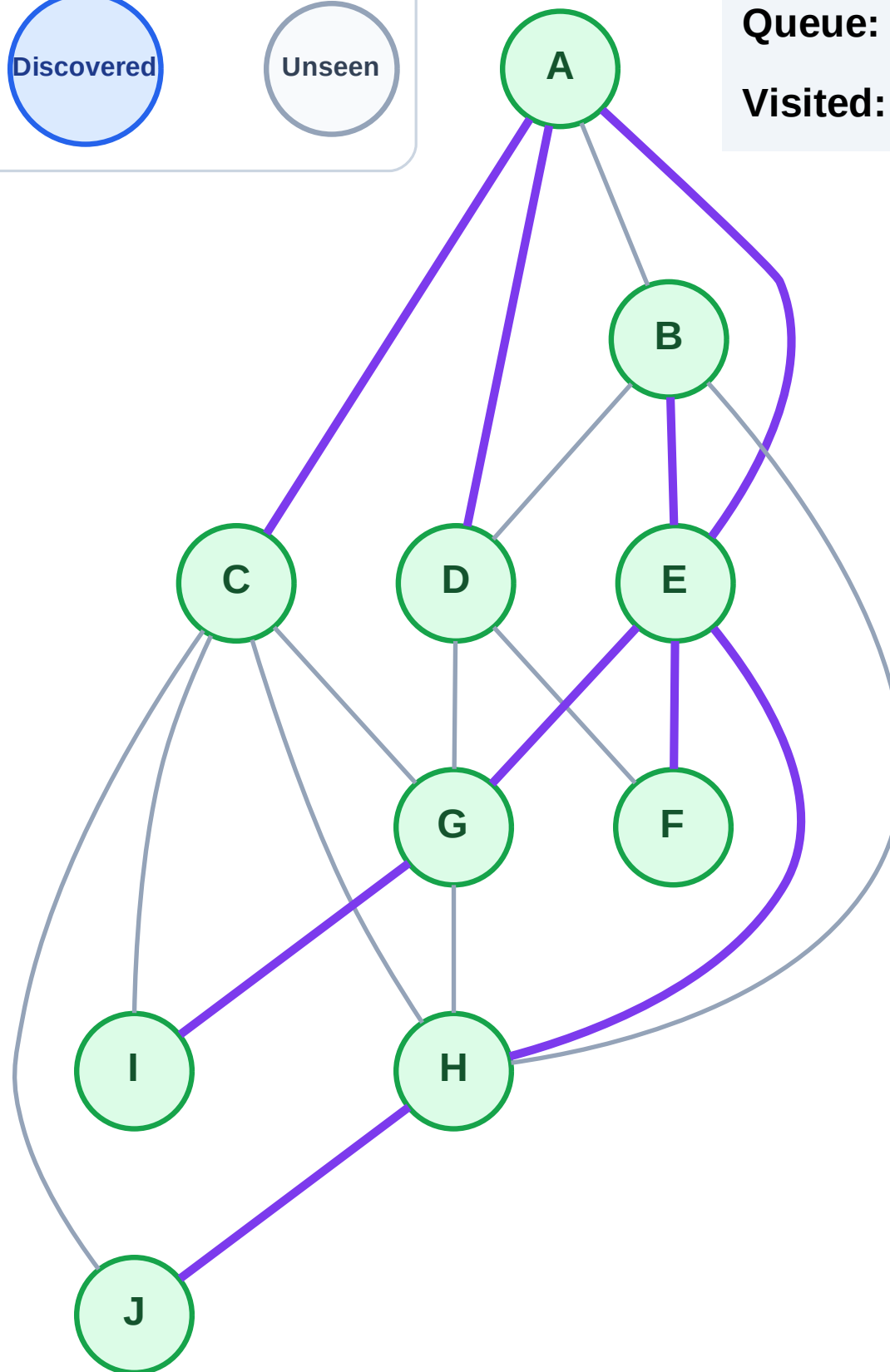
Step 21: No new neighbors from J

Legend



Queue: (empty)

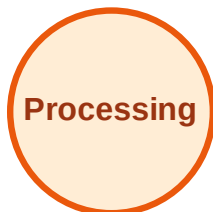
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

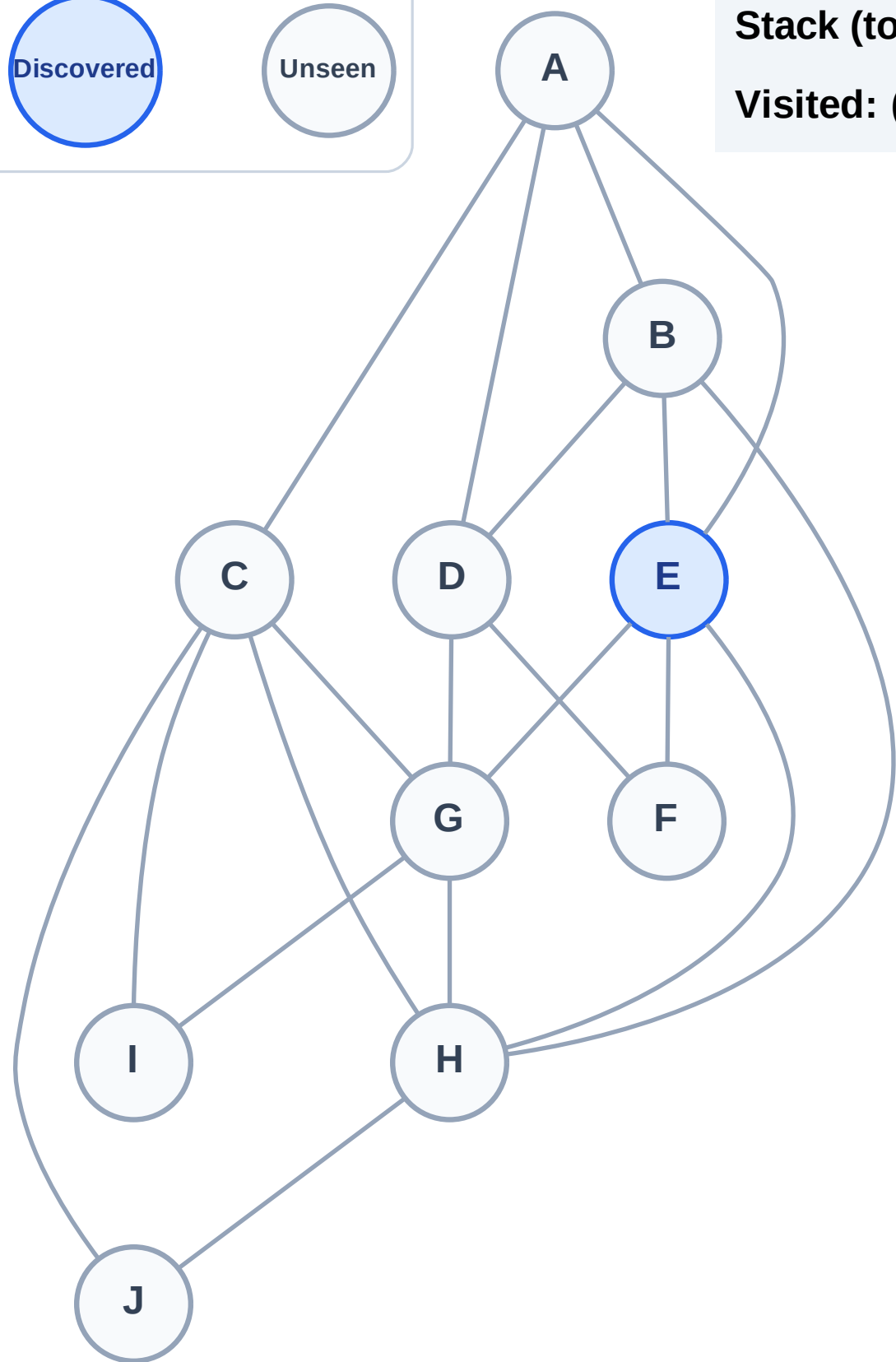
Step 1: Start at E

Legend



Stack (top at right): E

Visited: (none)



DFS Traversal

Step 2: Process node E

Legend

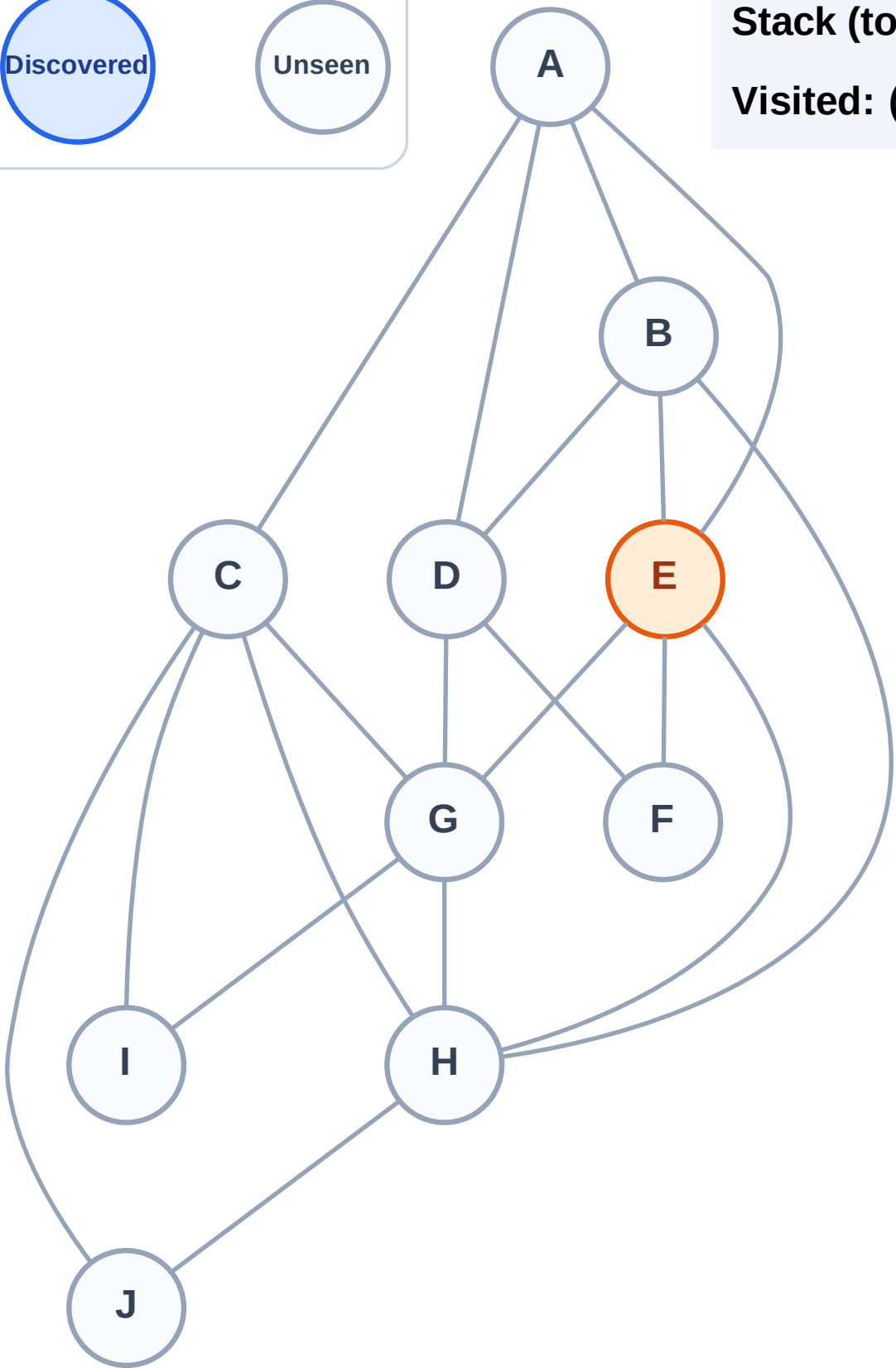
Complete

Processing

Discovered

Unseen

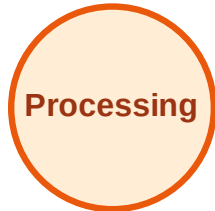
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

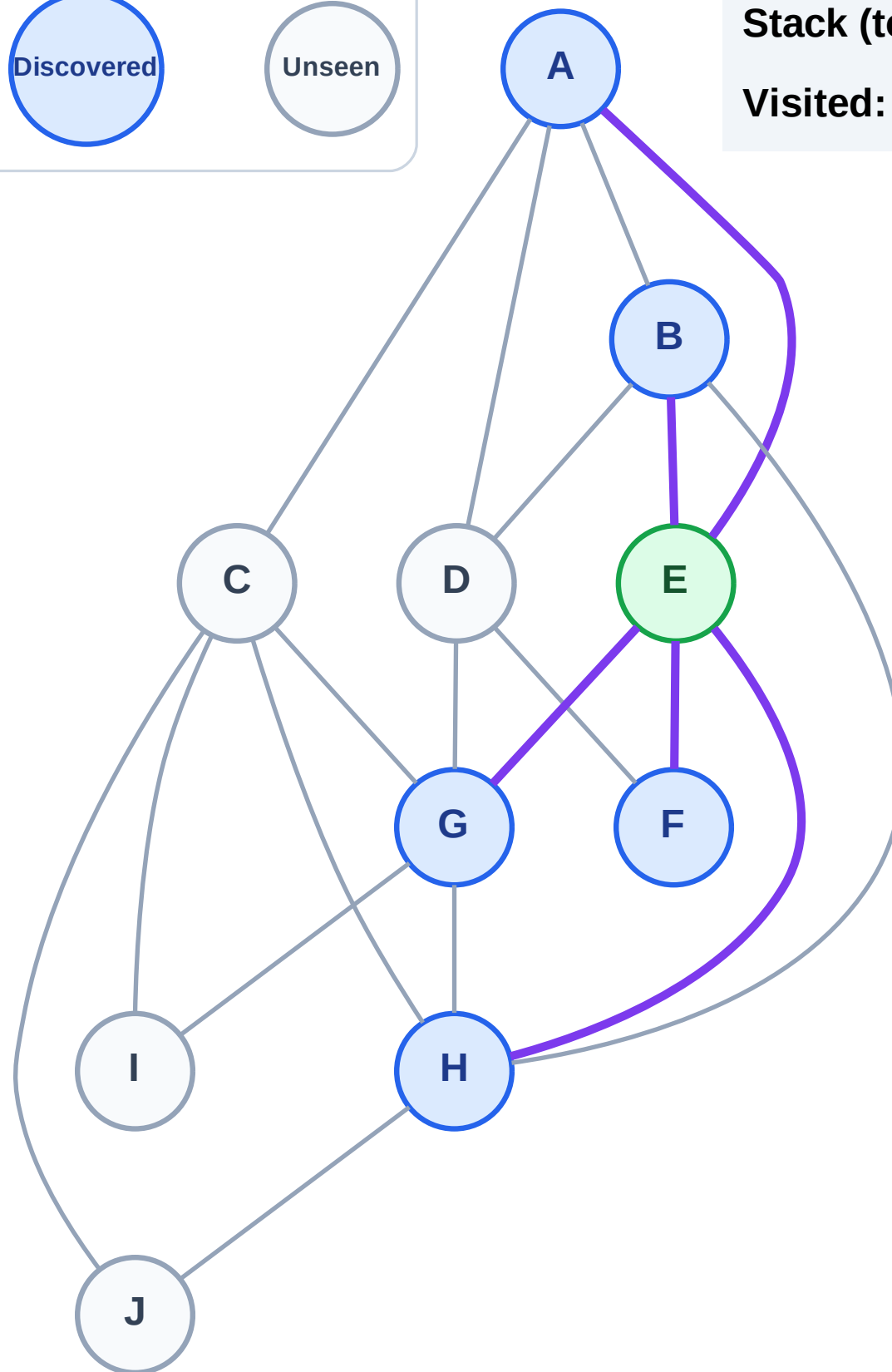
Step 3: Discover H, G, F, B, A from E

Legend



Stack (top at right): H | G | F | B | A

Visited: E



DFS Traversal

Step 4: Process node A

Legend

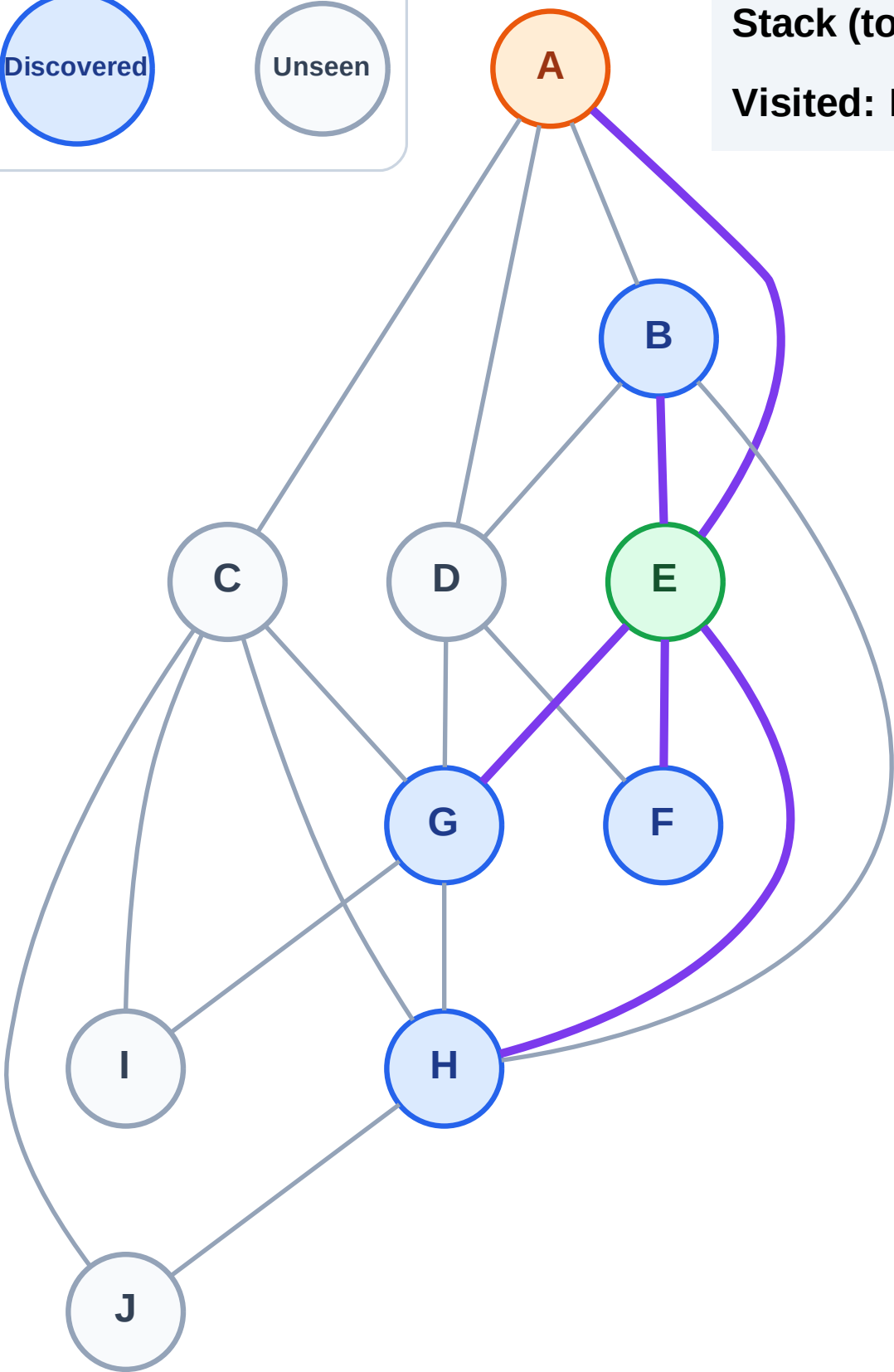
Complete

Processing

Discovered

Unseen

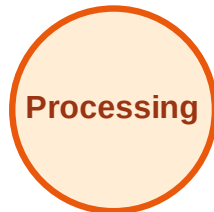
Stack (top at right): H | G | F | B
Visited: E



DFS Traversal

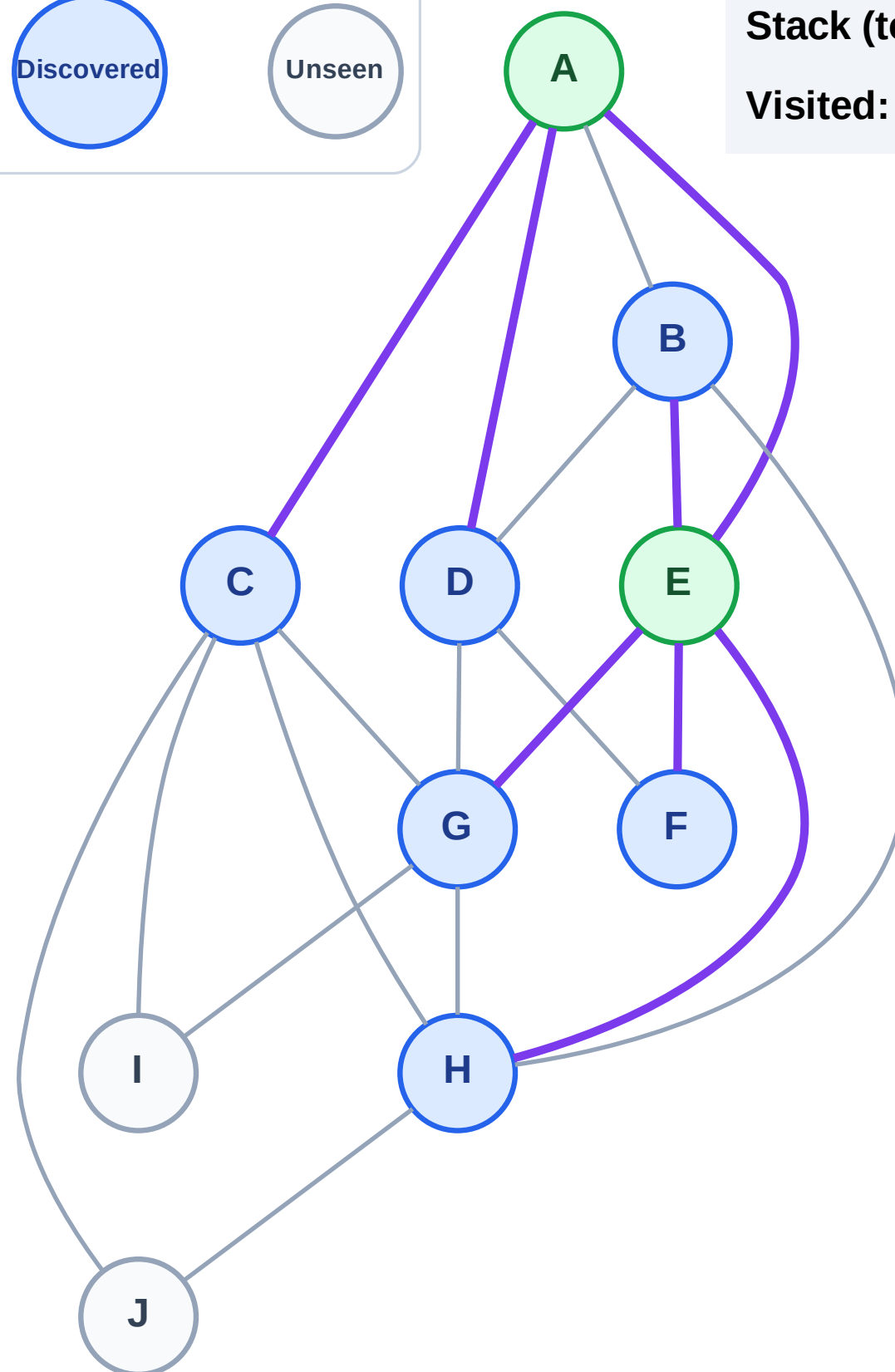
Step 5: Discover D, C from A

Legend



Stack (top at right): H | G | F | B | D | C

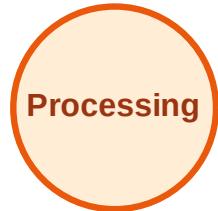
Visited: A, E



DFS Traversal

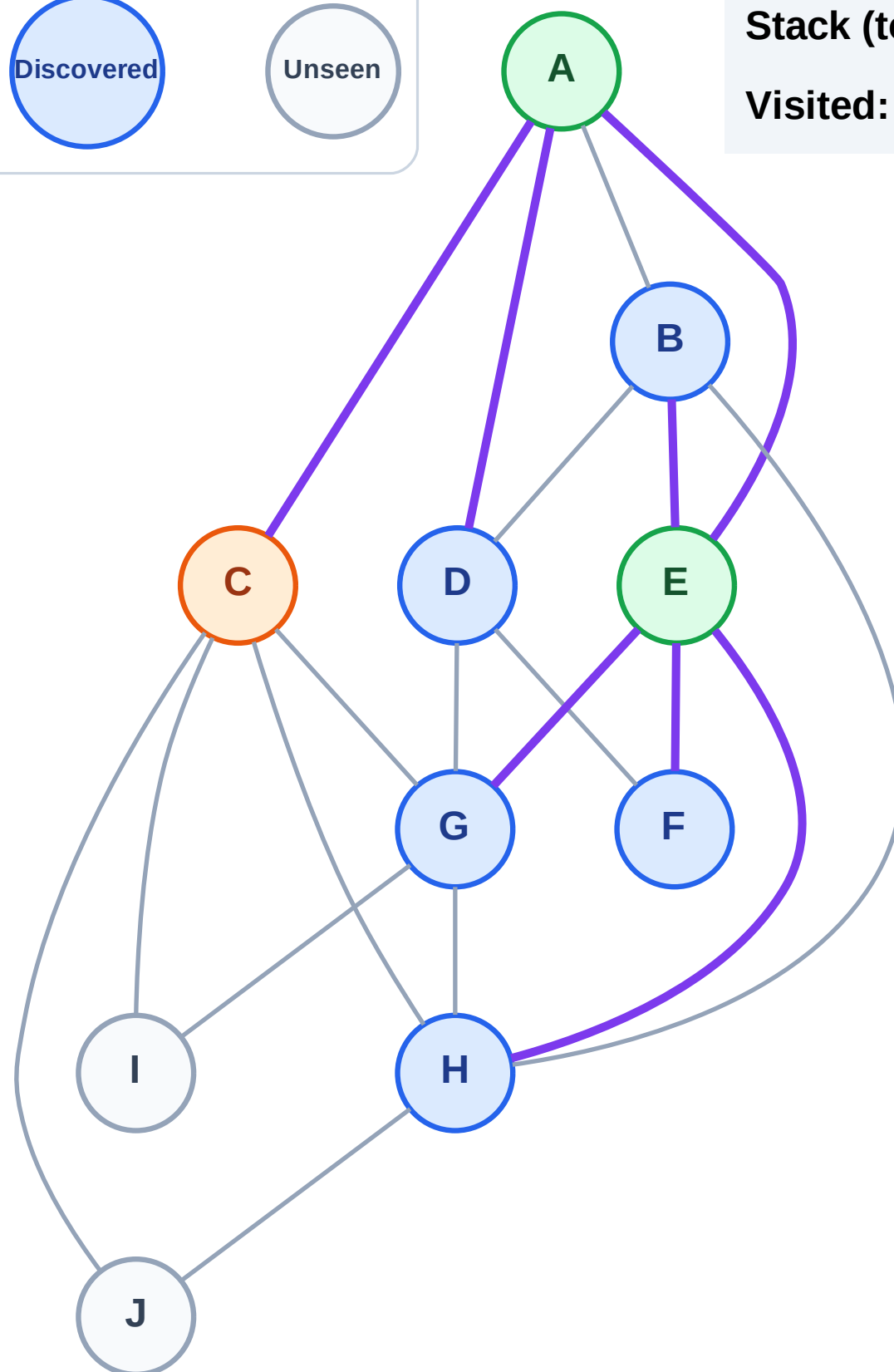
Step 6: Process node C

Legend



Stack (top at right): H | G | F | B | D

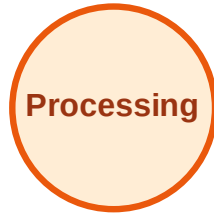
Visited: A, E



DFS Traversal

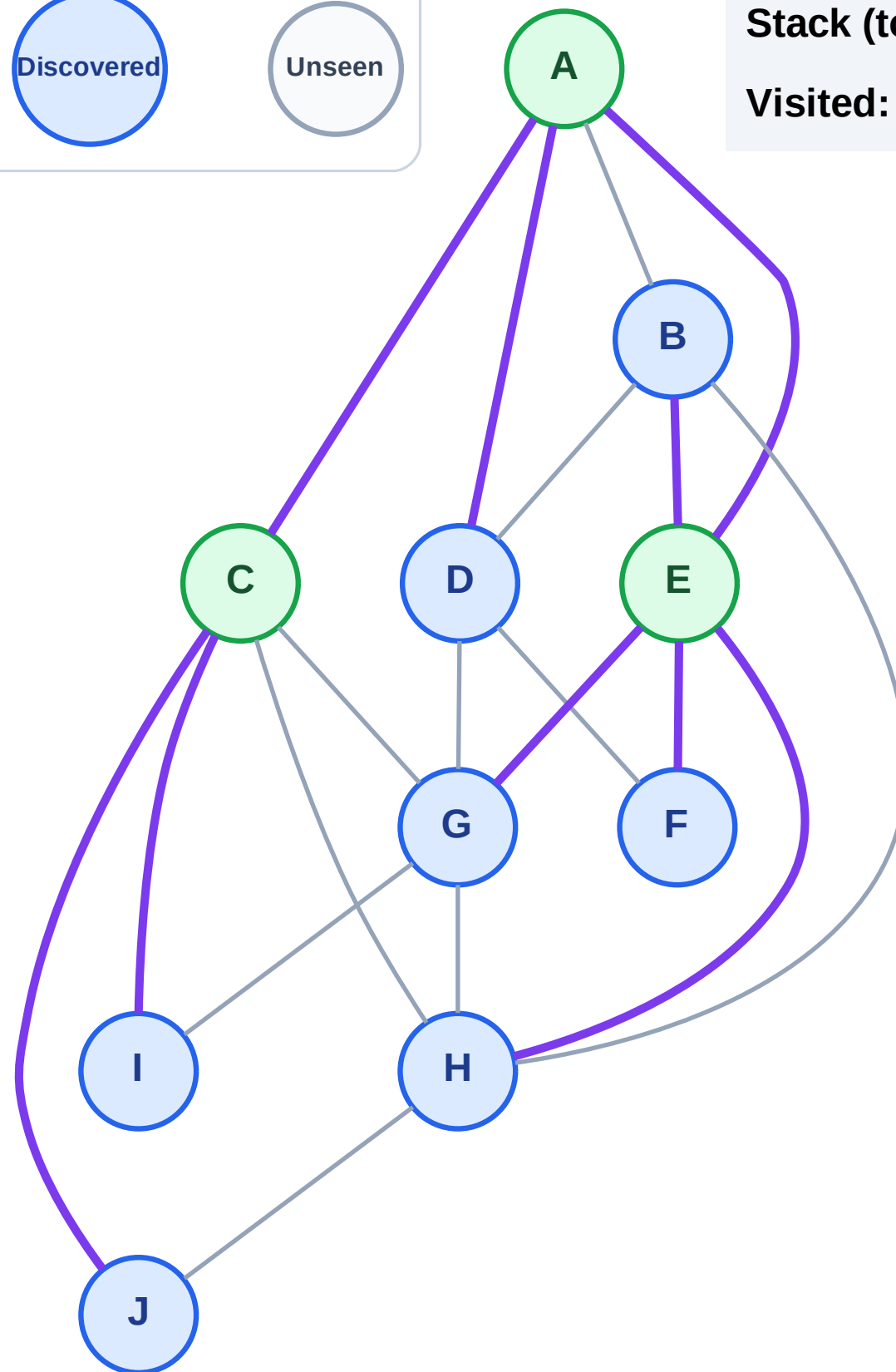
Step 7: Discover J, I from C

Legend



Stack (top at right): H | G | F | B | D | J | I

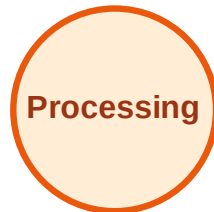
Visited: A, C, E



DFS Traversal

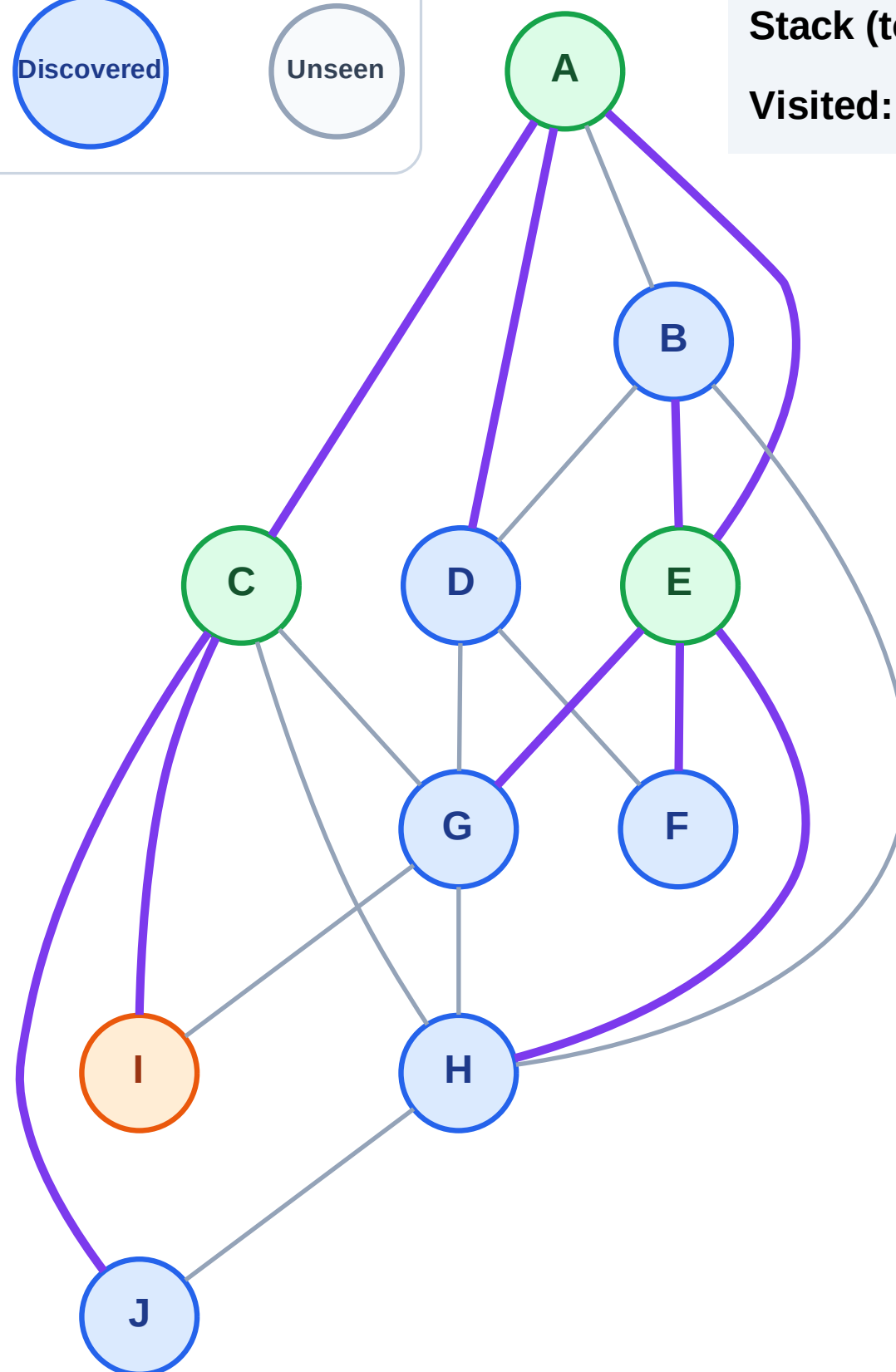
Step 8: Process node I

Legend



Stack (top at right): H | G | F | B | D | J

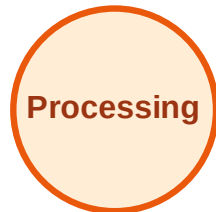
Visited: A, C, E



DFS Traversal

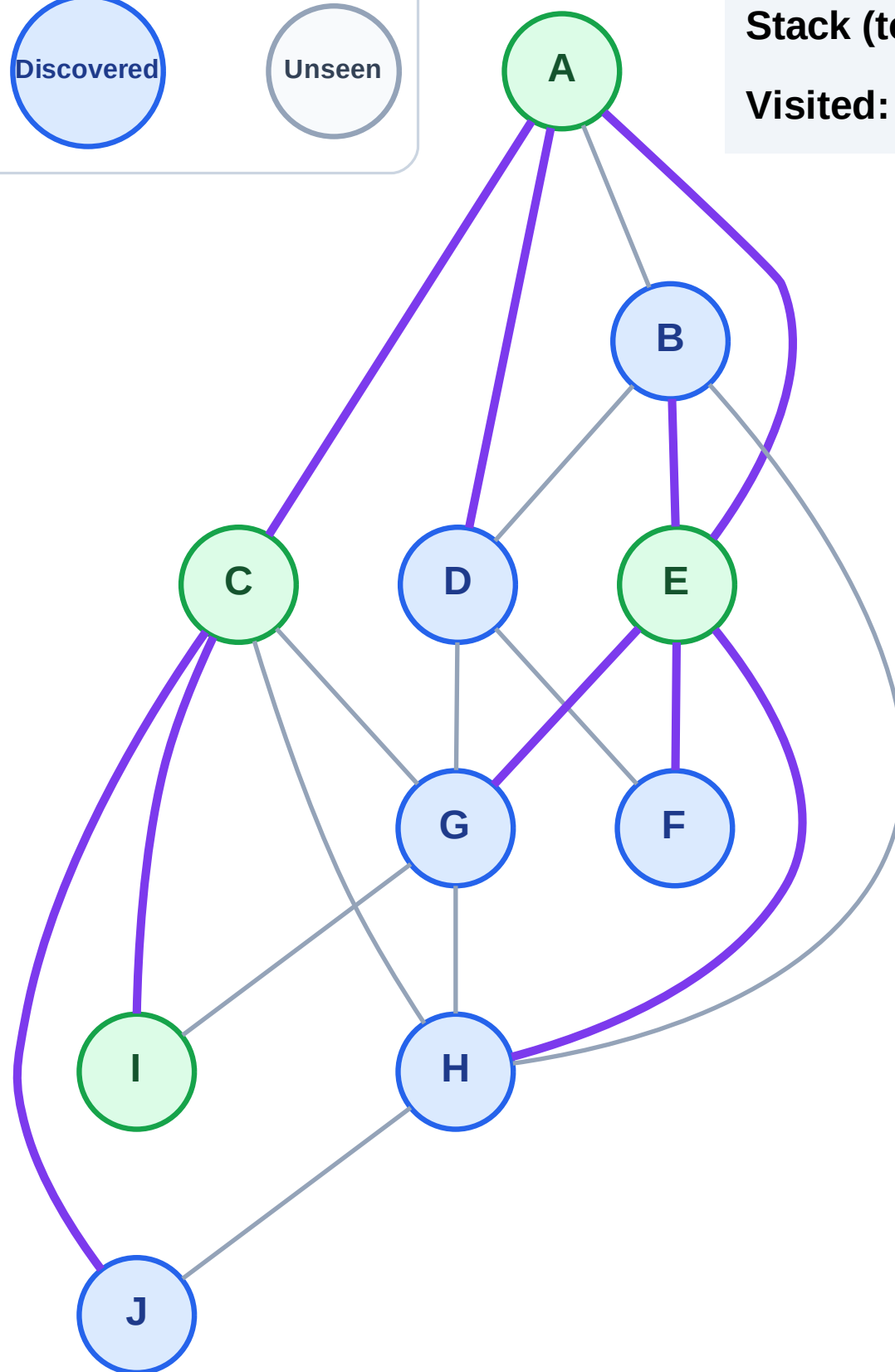
Step 9: No new neighbors from I

Legend



Stack (top at right): H | G | F | B | D | J

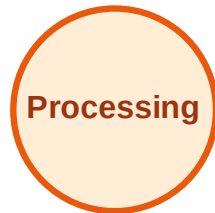
Visited: A, C, E, I



DFS Traversal

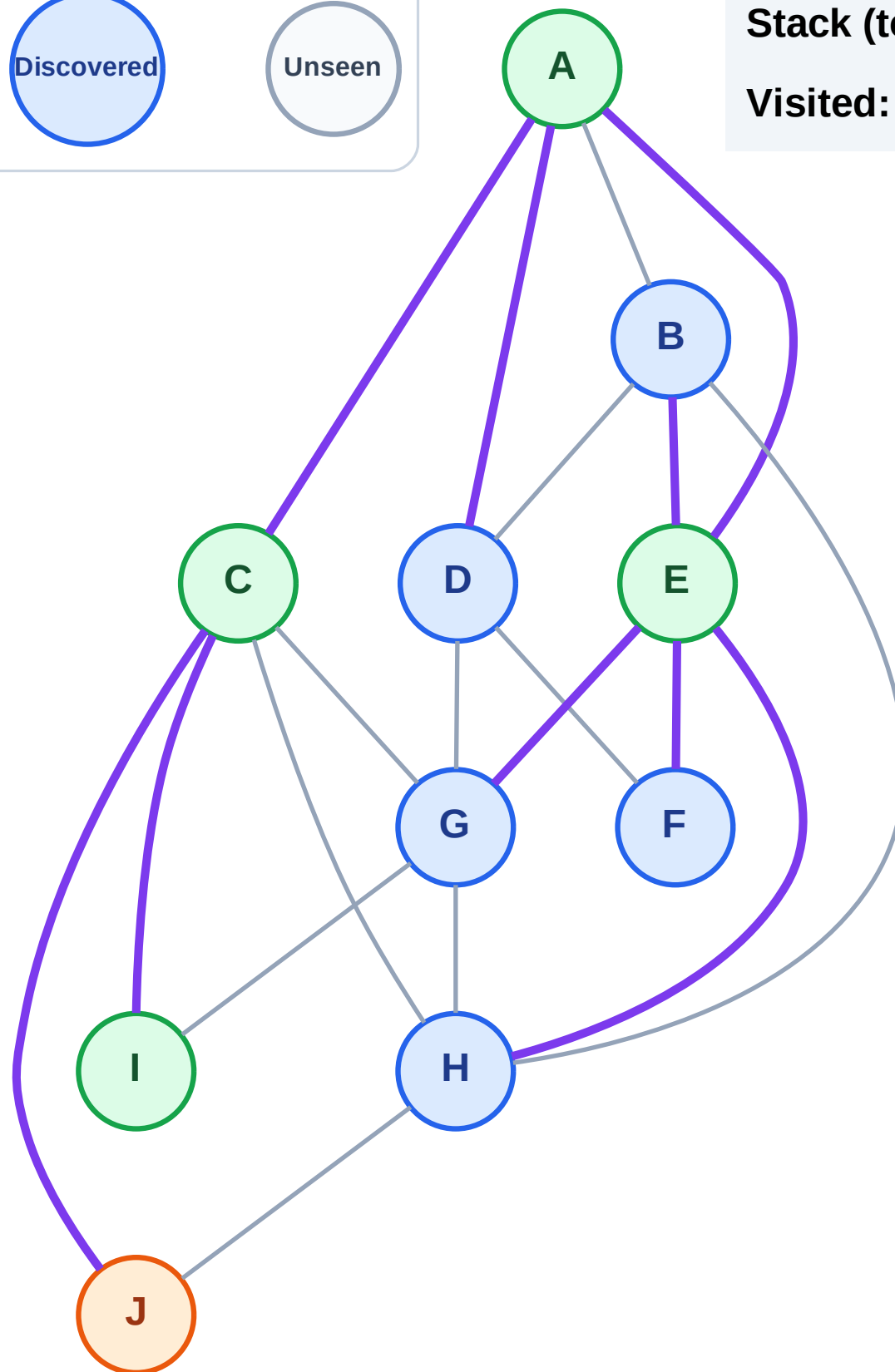
Step 10: Process node J

Legend



Stack (top at right): H | G | F | B | D

Visited: A, C, E, I

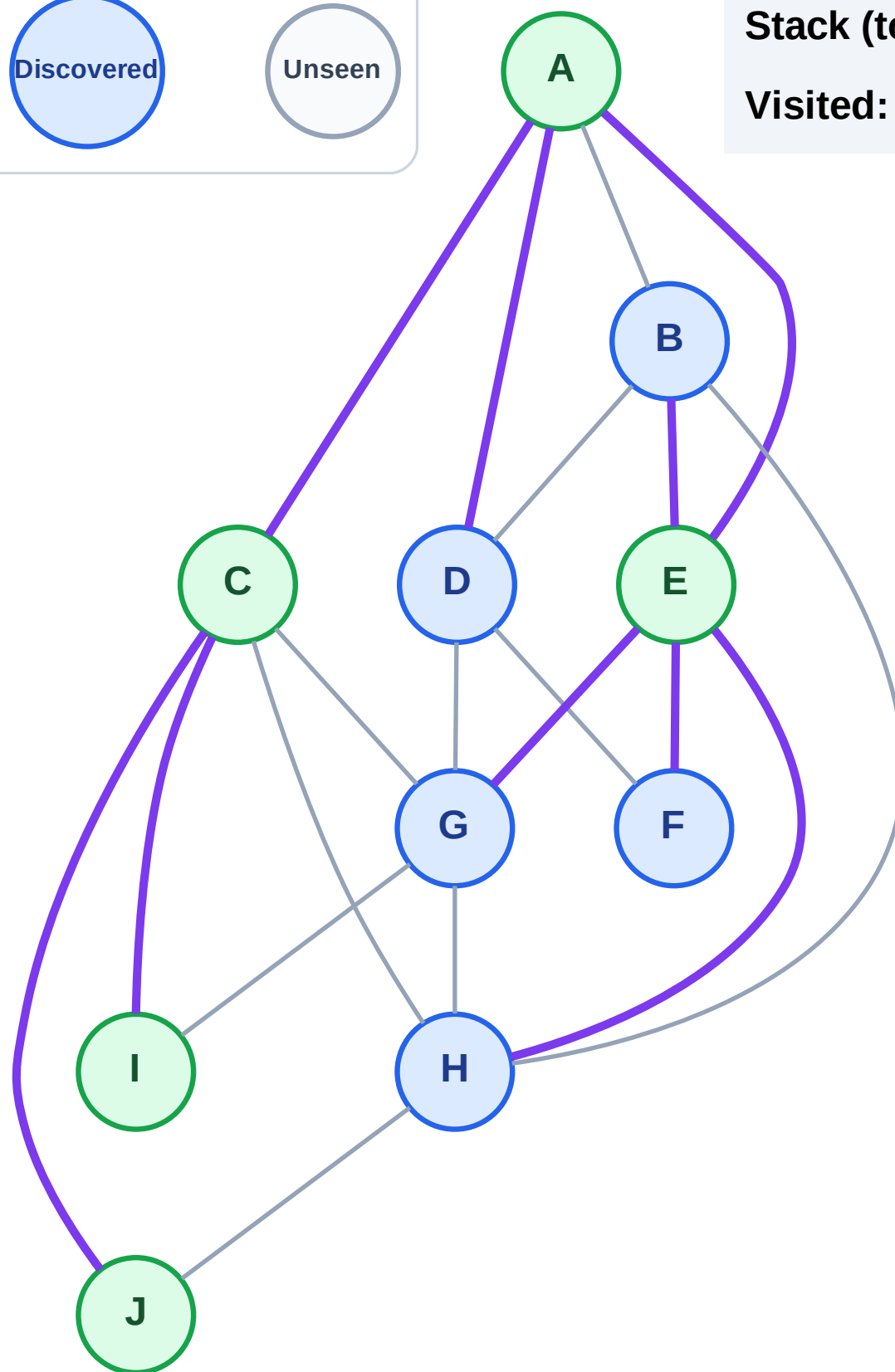


Step 11: No new neighbors from J

Legend

Unseen

Visited: A, C, E, I, J



DFS Traversal

Step 12: Process node D

Legend

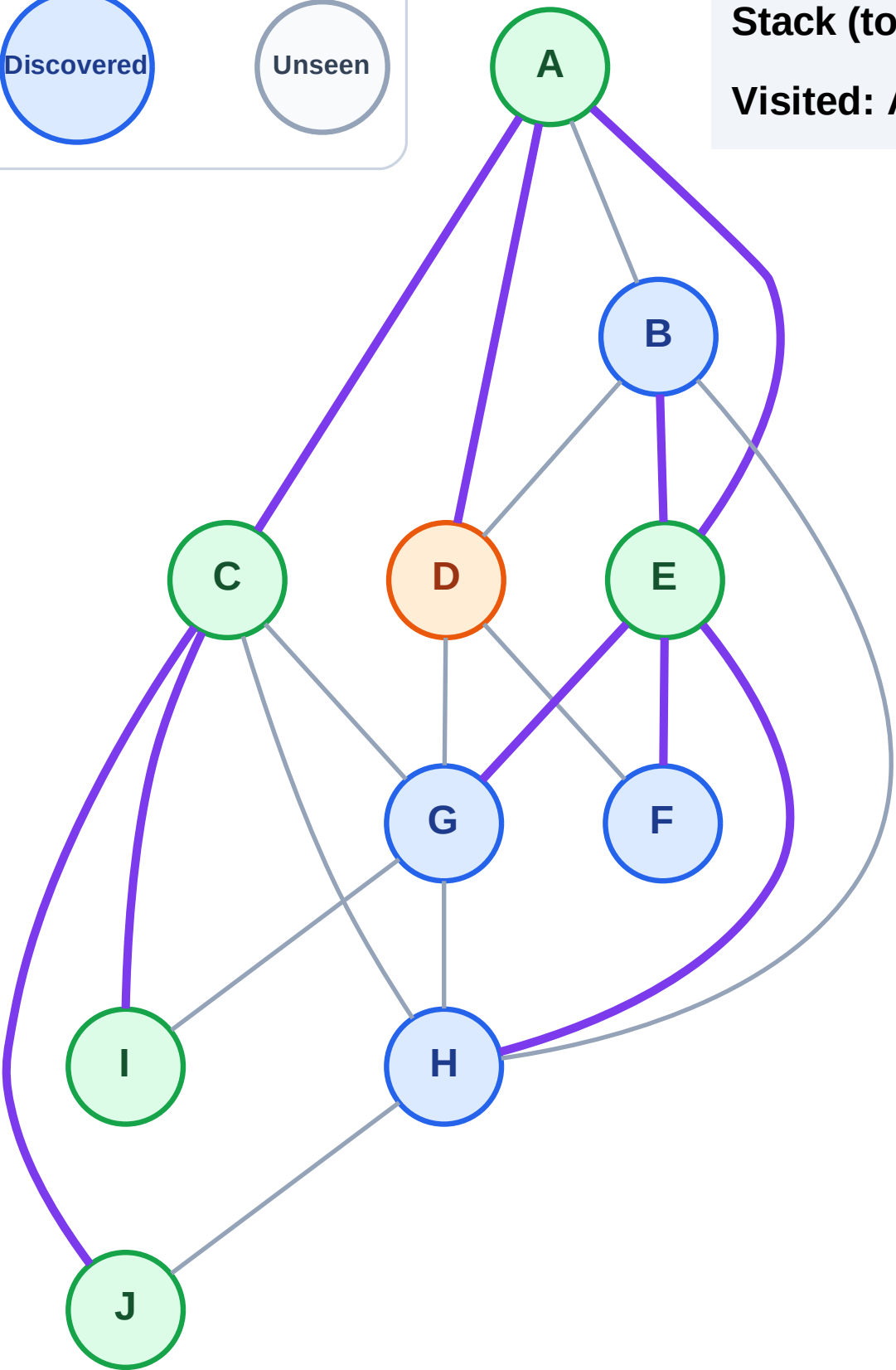
Complete

Processing

Discovered

Unseen

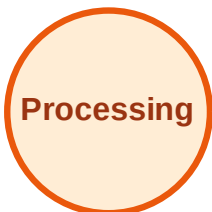
Stack (top at right): H | G | F | B
Visited: A, C, E, I, J



DFS Traversal

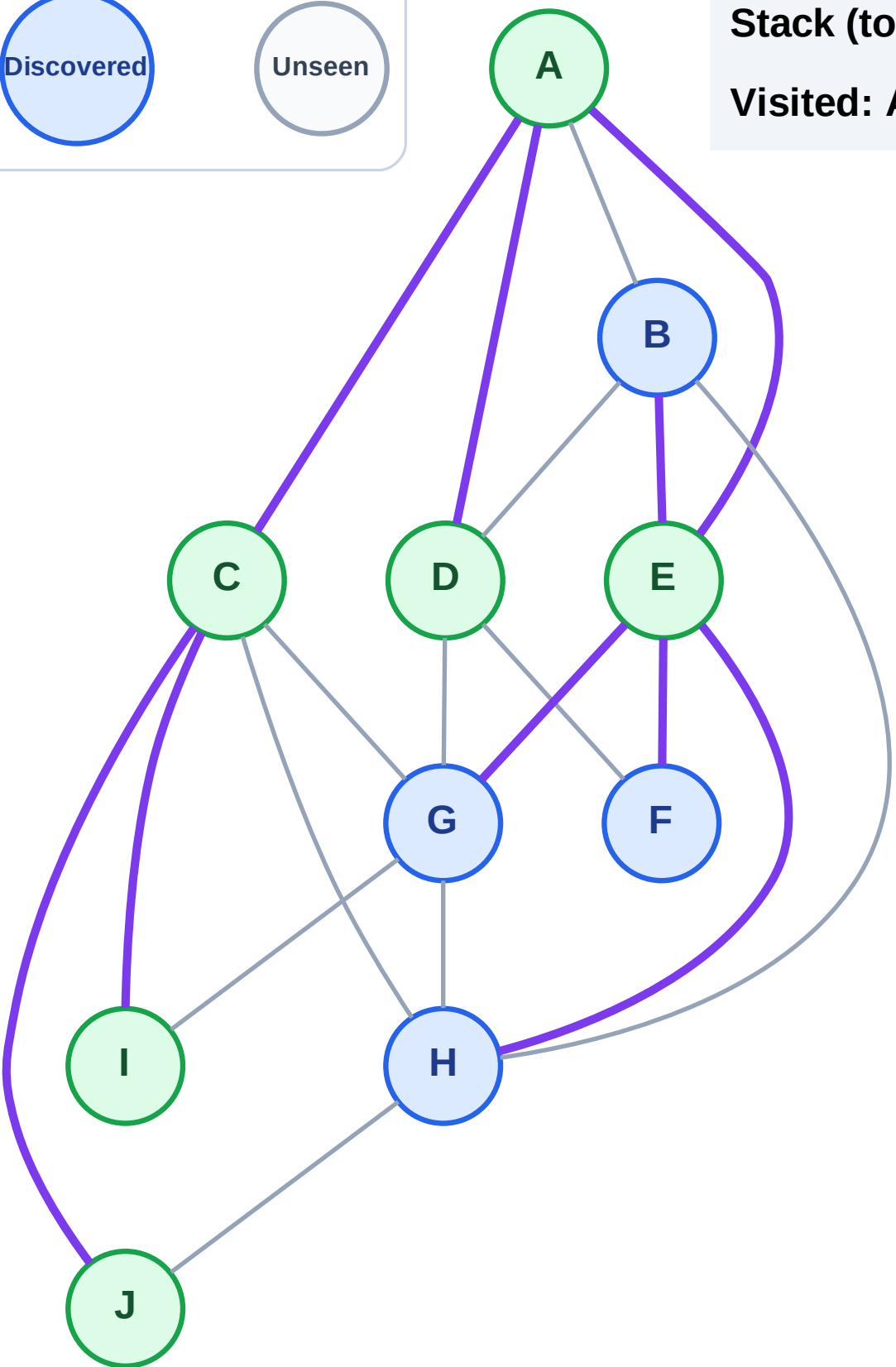
Step 13: No new neighbors from D

Legend



Stack (top at right): H | G | F | B

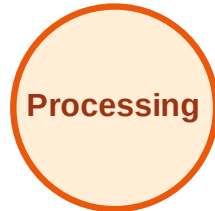
Visited: A, C, D, E, I, J



DFS Traversal

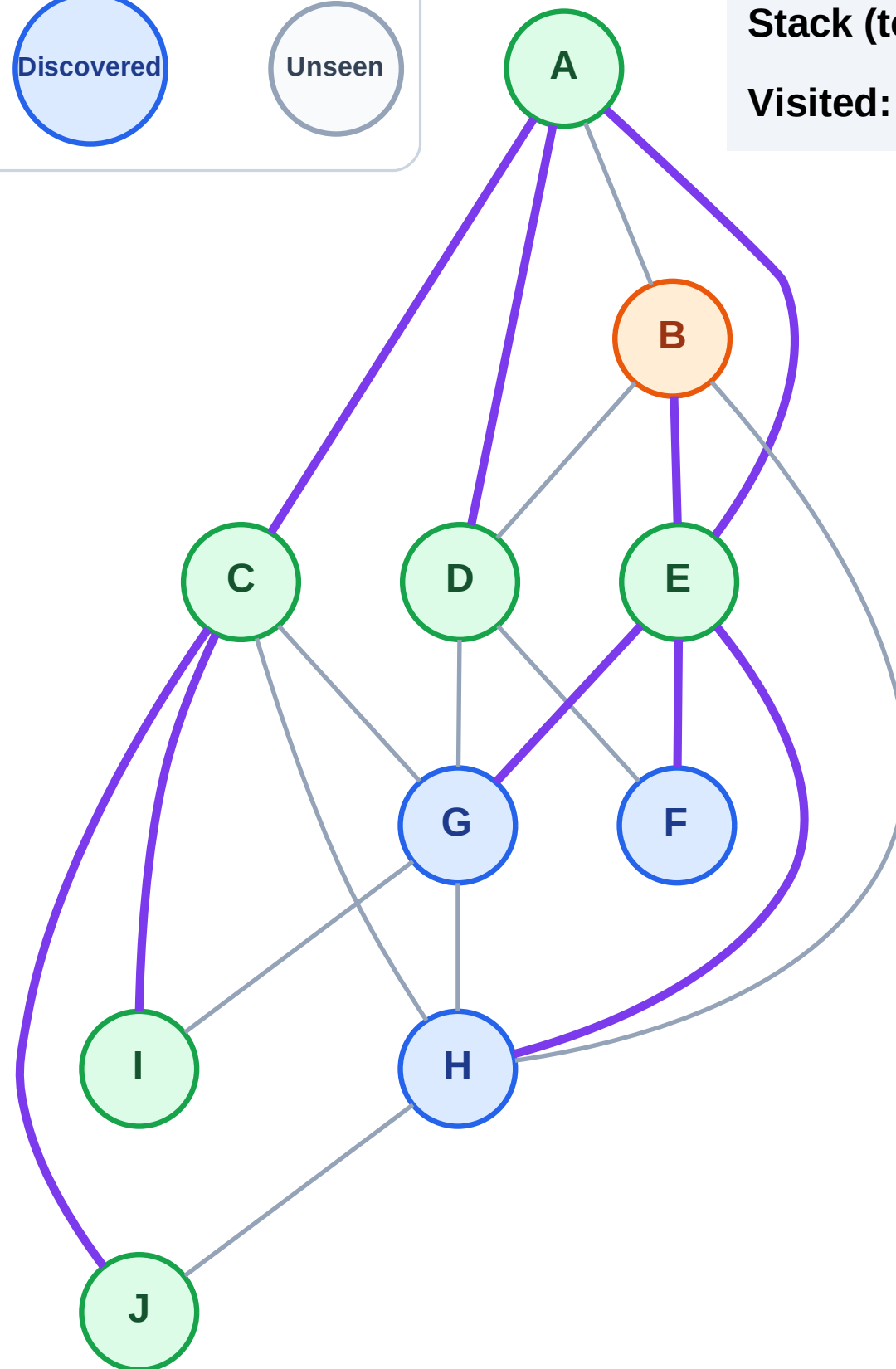
Step 14: Process node B

Legend



Stack (top at right): H | G | F

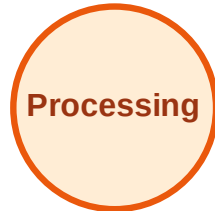
Visited: A, C, D, E, I, J



DFS Traversal

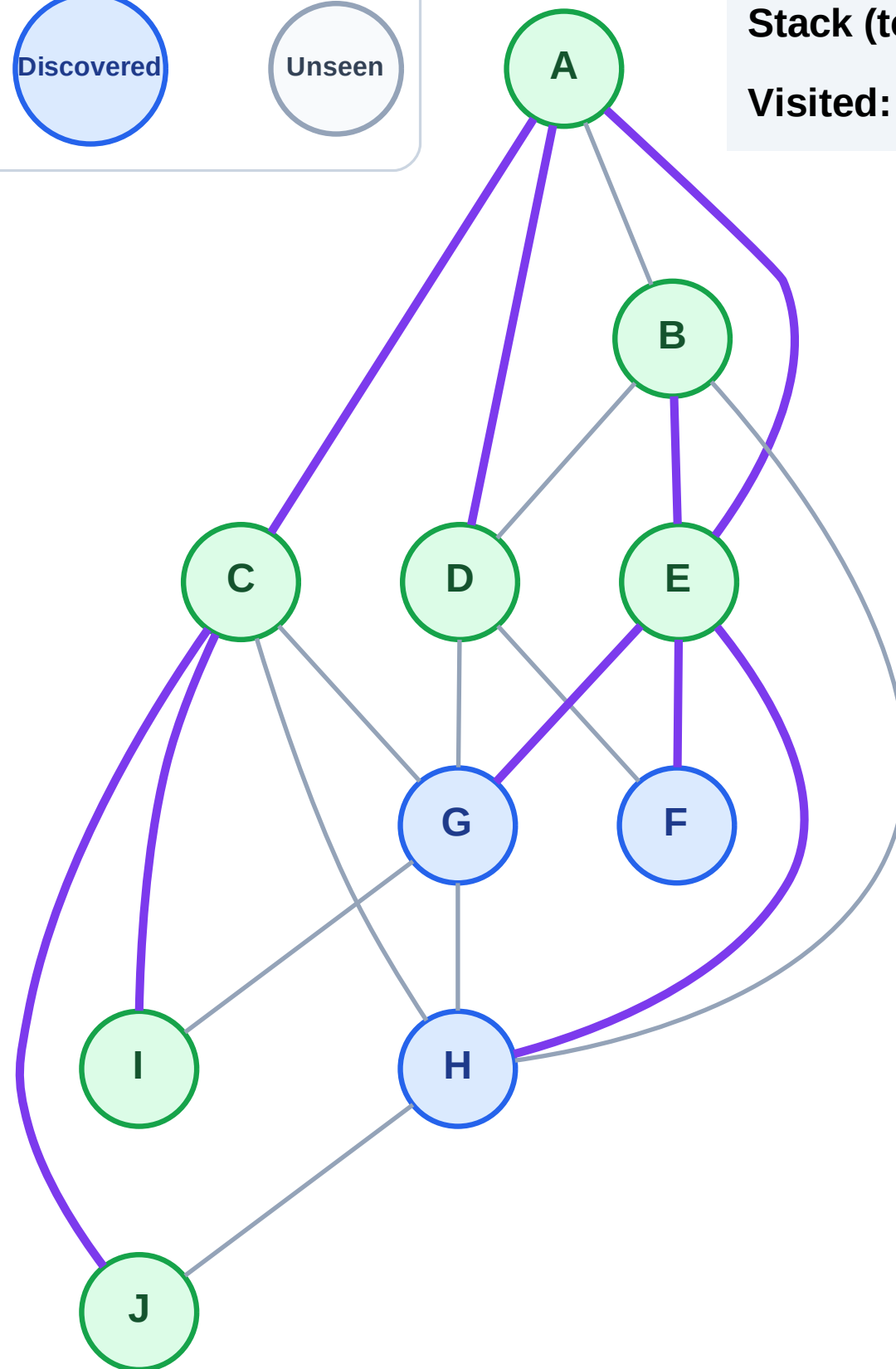
Step 15: No new neighbors from B

Legend



Stack (top at right): H | G | F

Visited: A, B, C, D, E, I, J



DFS Traversal

Step 16: Process node F

Legend

Complete

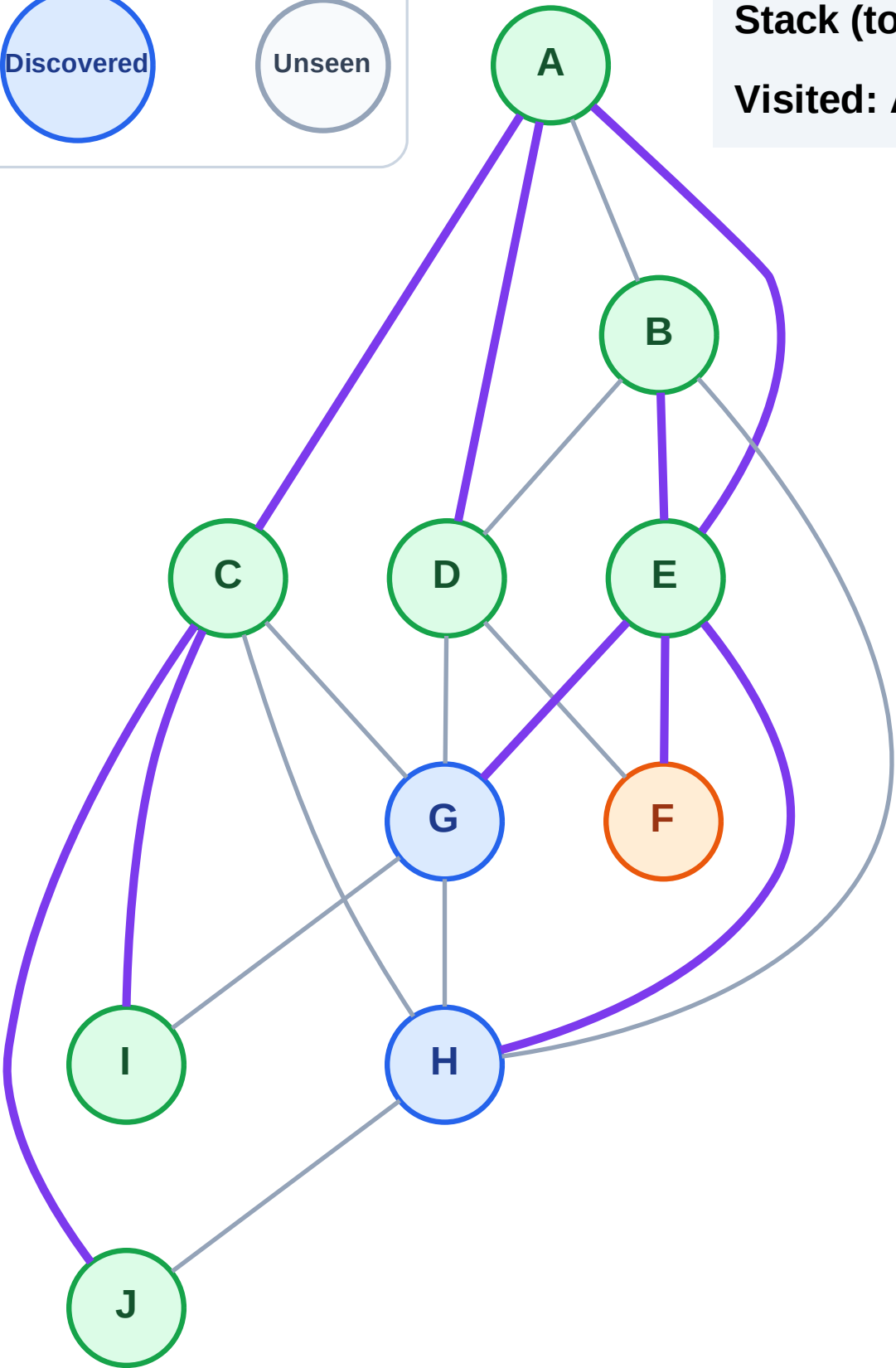
Processing

Discovered

Unseen

Stack (top at right): H | G

Visited: A, B, C, D, E, I, J



DFS Traversal

Step 17: No new neighbors from F

Legend

Complete

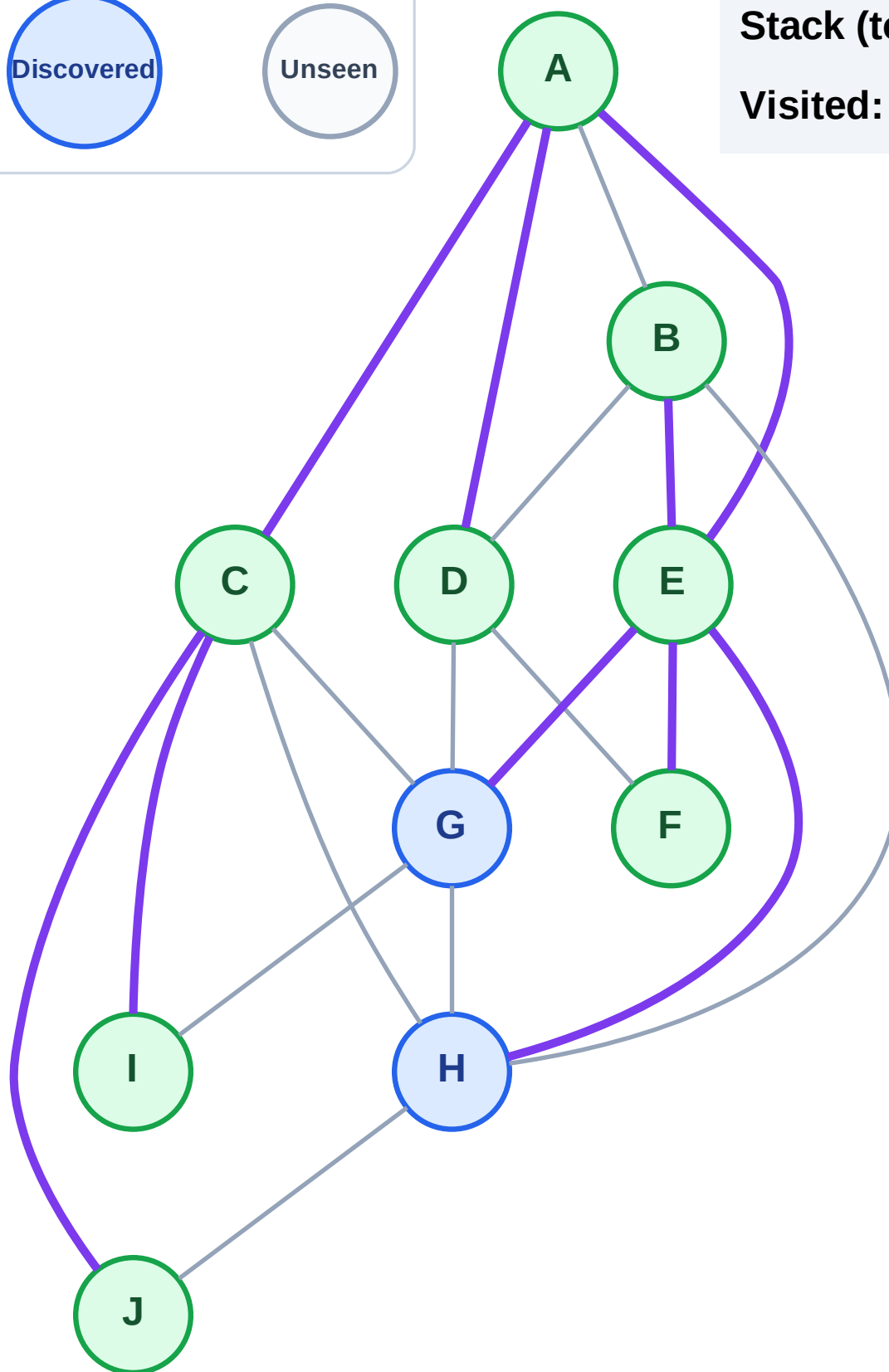
Processing

Discovered

Unseen

Stack (top at right): H | G

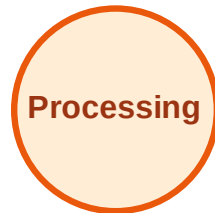
Visited: A, B, C, D, E, F, I, J



DFS Traversal

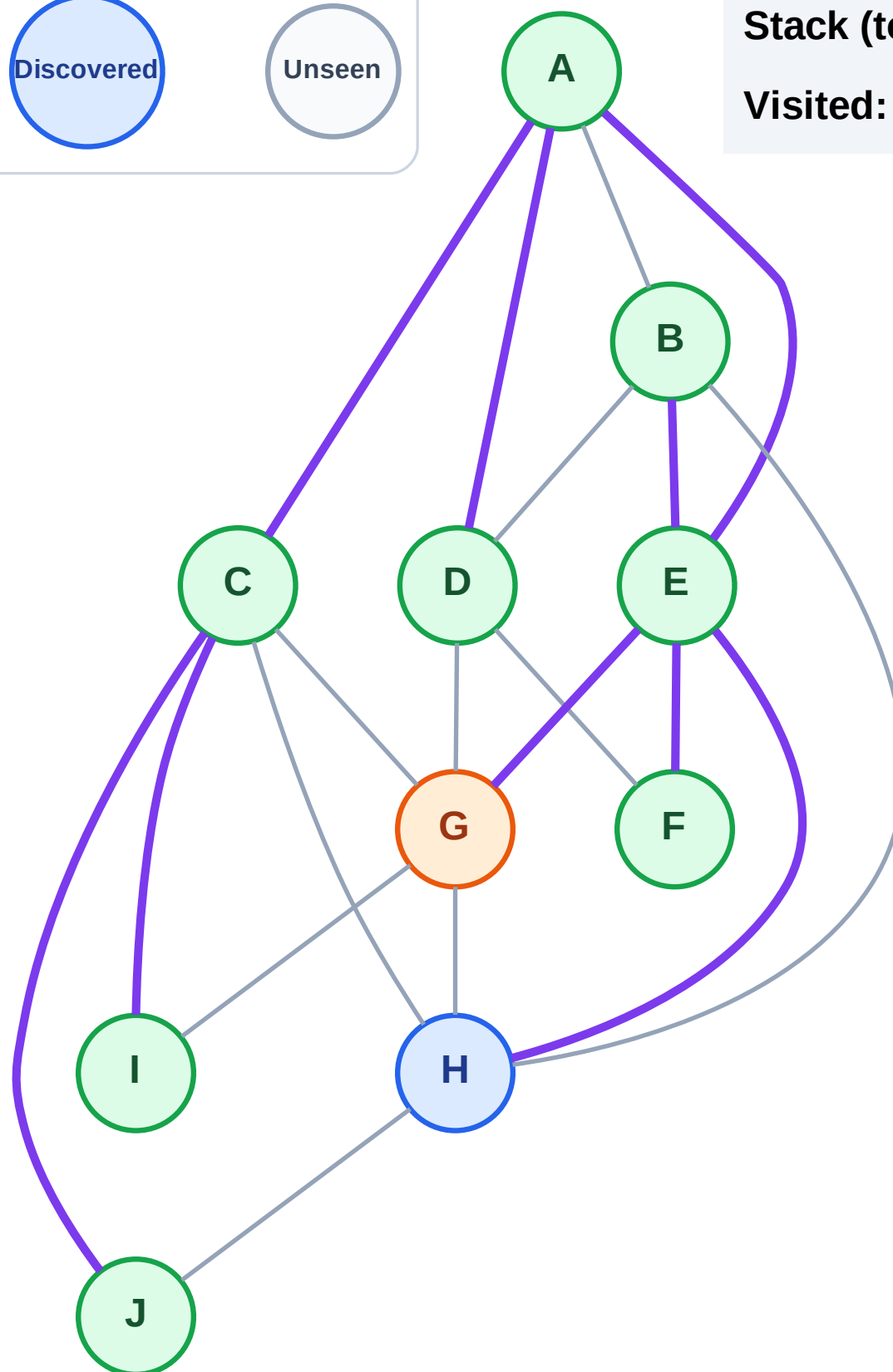
Step 18: Process node G

Legend



Stack (top at right): H

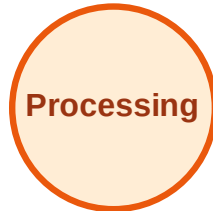
Visited: A, B, C, D, E, F, I, J



DFS Traversal

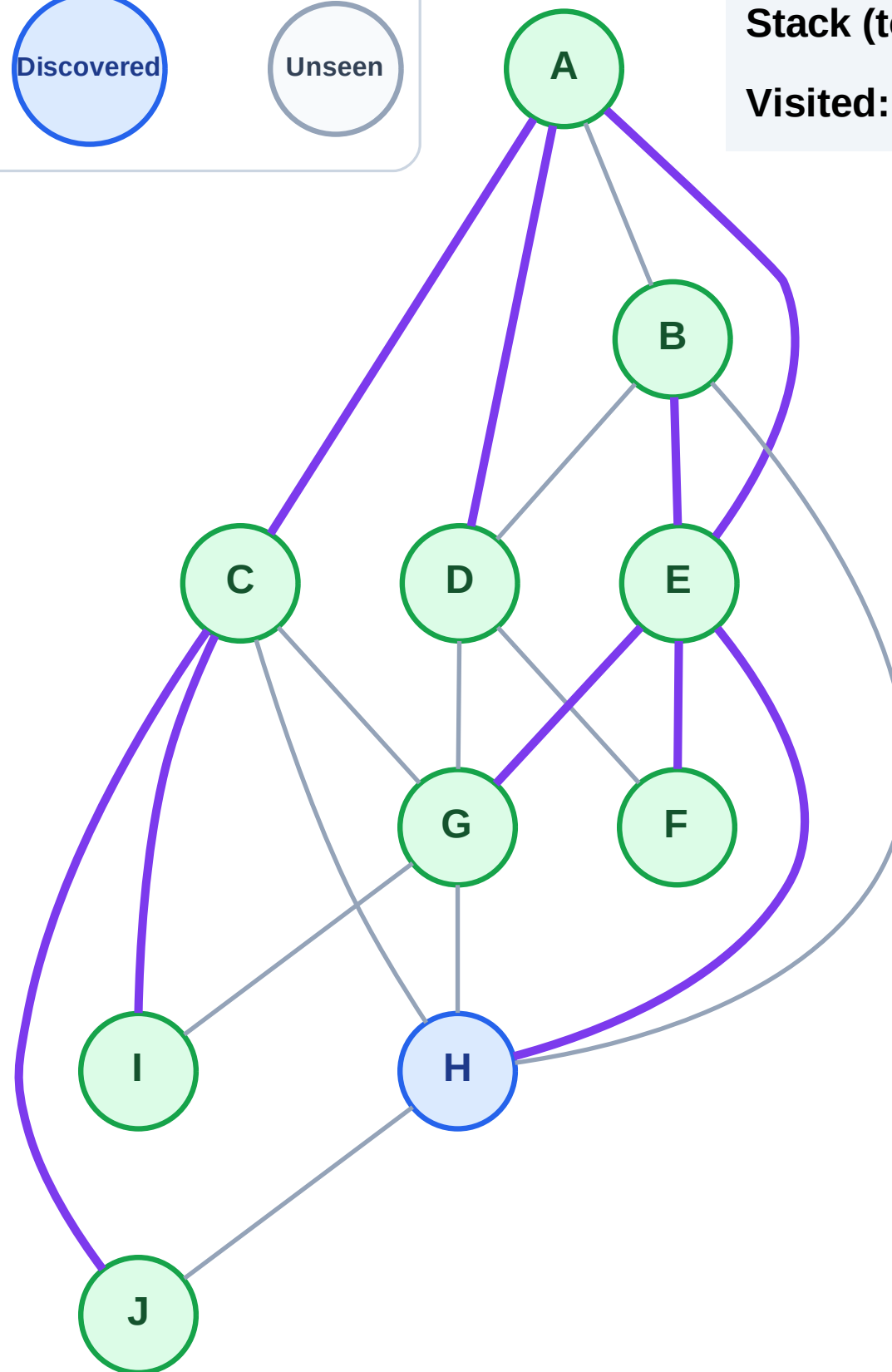
Step 19: No new neighbors from G

Legend



Stack (top at right): H

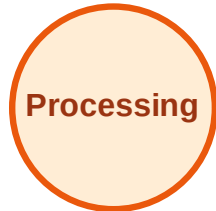
Visited: A, B, C, D, E, F, G, I, J



DFS Traversal

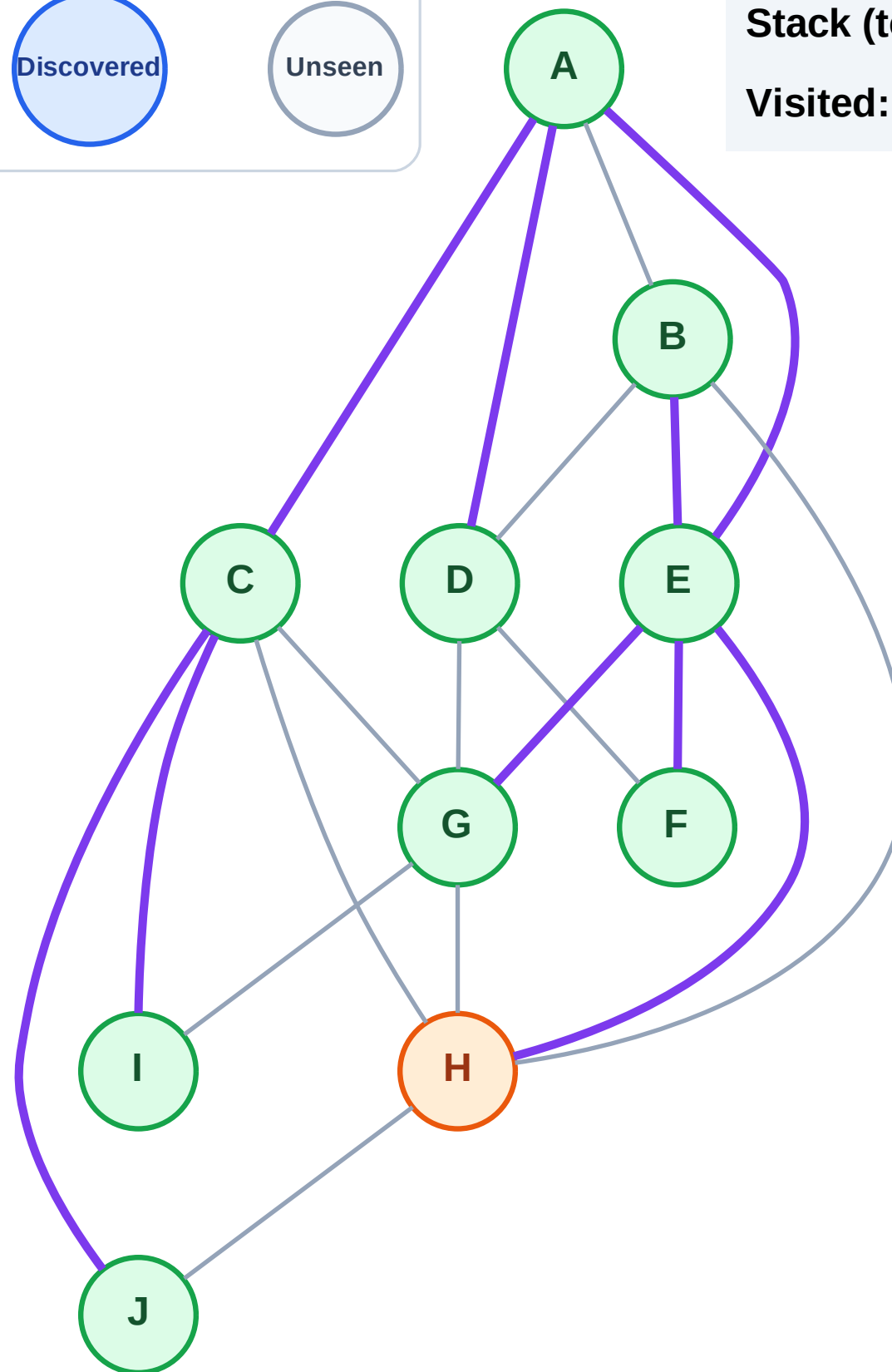
Step 20: Process node H

Legend



Stack (top at right): (empty)

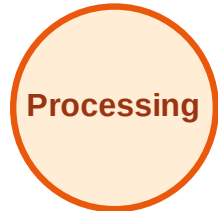
Visited: A, B, C, D, E, F, G, I, J



DFS Traversal

Step 21: No new neighbors from H

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

